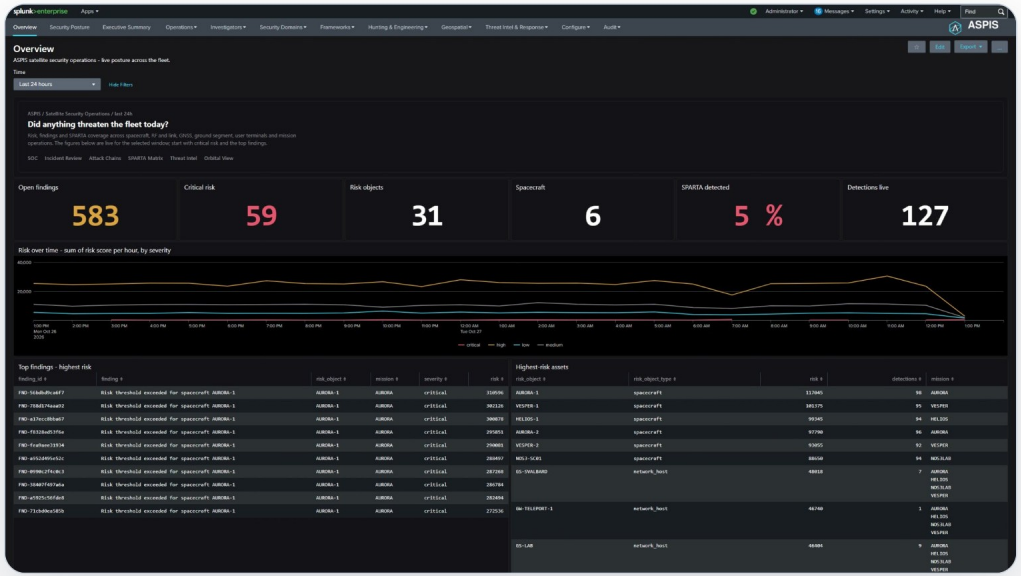
EXECUTIVE SUMMARY · PLATFORM WALKTHROUGH

ASPIS

Advanced Space Protection and Intelligence for Splunk

One place to see, score and work every cyber threat to a space mission, from the ground station to the spacecraft.

Installable in your current Splunk Enterprise Environment



EXECUTIVE SUMMARY · WHAT ASPIS IS

A security operations center for space missions, inside the Splunk you already run

01 · SEES

Every segment

Spacecraft and flight software, commanding, RF links, GNSS and timing, ground systems, user terminals and the supply chain.

02 · SCORES

Against SPARTA

127 detections mapped to SPARTA, MITRE ATT&CK, NIST CSF and NIST 800-53, with risk weighted by asset, mission and phase.

03 · WORKS

Like an ES SOC

Incident Review, investigations, suppressions, threat intel and response actions, each with an audit trail.

BOTTOM LINE The SOC finally sees the spacecraft, not just the ground network, with the tools and skills it already has. ASPIS is installable in your current Splunk Enterprise Environment.

EXECUTIVE SUMMARY · WHY IT MATTERS

Space attacks don’t look like IT attacks, and the impact is the mission

The challenge	What ASPIS does	The outcome
Space telemetry isn’t IT telemetry	Reads CCSDS frames, flight-software events, command logs, RF and GNSS metrics natively	Visibility where generic SOC content is blind
Attacks cross segments	Links ground, link and space detections into ordered attack chains	One campaign to work, not fifty unrelated alerts
Impact is mission impact	Weights every risk score by asset criticality, mission criticality and mission phase	The riskiest thing to the mission gets worked first
You have to prove it	Maps every detection to NIST CSF and 800-53, measures SPARTA coverage and audits every triage change	Compliance evidence comes from the same system

EXECUTIVE SUMMARY · DATA IN

Getting data in: the Splunk ingestion you already run, nothing new to deploy

MISSION SOURCES

- Telemetry and flight software
- Commanding: YAMCS, COSMOS
- RF and link security (SDR)
- GNSS and timing receivers
- Ground IT: logins, network
- Supply chain: SBOM, firmware

COLLECT

ASPIS adapters → HEC

CCSDS decoder, YAMCS poller, SDR bridge, GNSS and CubeSatSim readers, MQTT, Kafka or UDP bridge

Standard Splunk inputs

Universal Forwarder file monitors, syslog and UDP for anything that already writes logs

SPLUNK ENTERPRISE

6 telemetry indexes

spacecraft, command, rf, nav, ground and supply

127 detections

read the normalized data every 10 minutes and score what they find

WHAT YOU GET

Nothing new to deploy

Uses the HEC and forwarders you already run; inputs ship disabled.

Data stays where it lives

Index macros point ASPIS at your existing indexes; nothing moves.

Lab to flight without rework

Simulators and live missions share sourcetypes; detections don’t change.

EXECUTIVE SUMMARY · CIM COMPLIANCE

Making it CIM compliant: one set of field names from the ground to orbit

RAW FORMATS

- cFE event messages (text)
- YAMCS and COSMOS command logs
- NMEA sentences from GNSS
- SDR measurements (CSV, JSON)
- Ground logins and network flows
- SBOM and integrity records

NORMALIZE

At ingest: ASPIS JSON

Adapters emit shared keys: mission, spacecraft, segment, subsystem and component

At search: field mapping

Extractions, aliases, evals and lookups map vendor fields to the same names

MODEL

Ground IT → Splunk CIM

Tagged into the Authentication, Network Traffic and Change data models

Space data → SSCIM

Extends CIM, reusing user, src, dest, action, app, signature and severity

WHAT YOU GET

Fits existing CIM content

Ground data is tagged for the CIM data models your searches use.

Detections written once

Swap a telemetry source by changing the mapping, not the content.

One shared vocabulary

Analysts search every segment with the same field names.

WHAT SHIPS IN THE APP

ASPIS at a glance

33
consoles in the app menu

127
detections across 12 domains

6 + 6
risk rules and attack chains

22
hypothesis-driven hunts

9
indexes: 6 telemetry + risk, findings, summary

22
space and ground sourcetypes

17
KV collections of analyst state

184
saved searches in total

BEHIND THE SCENES

Five stages from raw telemetry to a closed case

1 · COLLECT

Adapters + HEC

CCSDS decoder, YAMCS poller, CubeSatSim, GNSS NMEA, SDR and bus bridges, NOS3 file inputs

2 · INDEX

6 telemetry indexes

spacecraft, command, rf, nav, ground, supply; 22 sourcetypes, CIM-tagged

3 · DETECT

127 detections

Scheduled every 10 min; each writes scored, framework-tagged risk events

4 · CORRELATE

12 finding rules

6 risk thresholds and 6 attack chains read satsec_risk and write findings

5 · TRIAGE

Consoles + response

Incident Review, Investigations, dashboards; outbox, intel and webhook actions

Analyst state lives in the KV store, not in the indexes:

`finding_reviews` `finding_review_audit` `finding_notes` `investigations` `investigation_notes` `suppressions` `threat_intel` `assets` `identities` `content_overrides` `ir_views` `+ 6 more`

BEHIND THE SCENES · DATA

Where the data lands

Index	What lands there	Key sourcetypes
satsec_spacecraft	Telemetry points and limit states; flight-software events	satsec:telemetry, satsec:fsw:event, nos3:cfs:evs
satsec_command	Telecommands with operator, source host and auth state	satsec:command, yamcs:cmdhist
satsec_rf	Link metrics (SNR, C/N0), emitters, SDLS alarms, ISL	satsec:rf, satsec:link:security
satsec_nav	GNSS fixes, C/N0, spoofing and jamming flags	satsec:nav
satsec_ground	Operator logins, ground network, contacts, terminals	satsec:ground:auth, satsec:contact
satsec_supply	Supplier and software-component events	satsec:supply
satsec_risk	Risk events written by the detections	stash (via collect)
satsec_findings	Findings from risk rules and attack chains	stash (via collect)
satsec_summary	Telemetry-health snapshots	stash (via collect)

BEHIND THE SCENES · DETECTION

One detection, end to end: SAT-CMD-001

1. Every 10 minutes, read telecommands from the last hour (−70 to −10 min)
2. Drop sources on the mission's command allowlist and known ground hosts
3. Tag it: SPARTA IA-0007.02, ATT&CK T0859, NIST DE.CM-01; base score 75, confidence 80
4. Fan out one risk event per risk object: the spacecraft and the source host
5. Weight the score, then write it to satsec_risk with collect

`satsec_risk_finalize`

```
risk = min(150,
base × confidence
× asset × mission
× phase)
```

Weights: criticality low 0.75 to critical 1.5; phase from 0.5 (simulation) to 1.5 (LEOP). A threat-intel match adds ×1.5.

75 × 0.8 = 60 on a nominal mission. On a critical spacecraft mid-maneuver it reaches the cap of 150.

BEHIND THE SCENES · CORRELATION

Risk becomes a finding in twelve ways

Risk-based rules

Rule	Raises a finding when
RBA-001	a high-fidelity detection fires (direct)
RBA-002	one object passes 100 risk in 24 h
RBA-003	one object shows 3+ SPARTA tactics in 7 d
RBA-004	5+ low or medium detections, no high, in 7 d
RBA-005	one mission passes 150 risk in 24 h
RBA-006	ground, link and space all hit one mission

Attack chains (ordered stages)

Chain	Story
CHN-001	Ground compromise to mission impact
CHN-002	Rogue ground station takeover
CHN-003	Supply-chain implant to onboard persistence
CHN-004	GNSS spoofing to navigation manipulation
CHN-005	Defense evasion, then hazardous commanding
CHN-006	Insider data theft

Findings are stamped when raised, keep a stable ID, honor suppressions, and are de-duplicated per object within a throttle window (1 hour by default).

BEHIND THE SCENES · TRIAGE

The same triage model as Enterprise Security

New → **In Progress** → **Pending** → **Resolved** → **Closed**

Groups: New · Open · Closed

Urgency

Severity × priority, where priority is the asset's criticality. Analysts may override it.

Disposition

Undetermined, true positive, benign positive, false positive (logic or data), other.

Audit

Every change appends a row: who, when, old to new. Comments need 20+ characters.

Writes go straight to the KV store through Splunk's REST API with the CSRF key, so the consoles never run risky SPL commands.

BEHIND THE SCENES · SCHEDULER

What runs, and how often

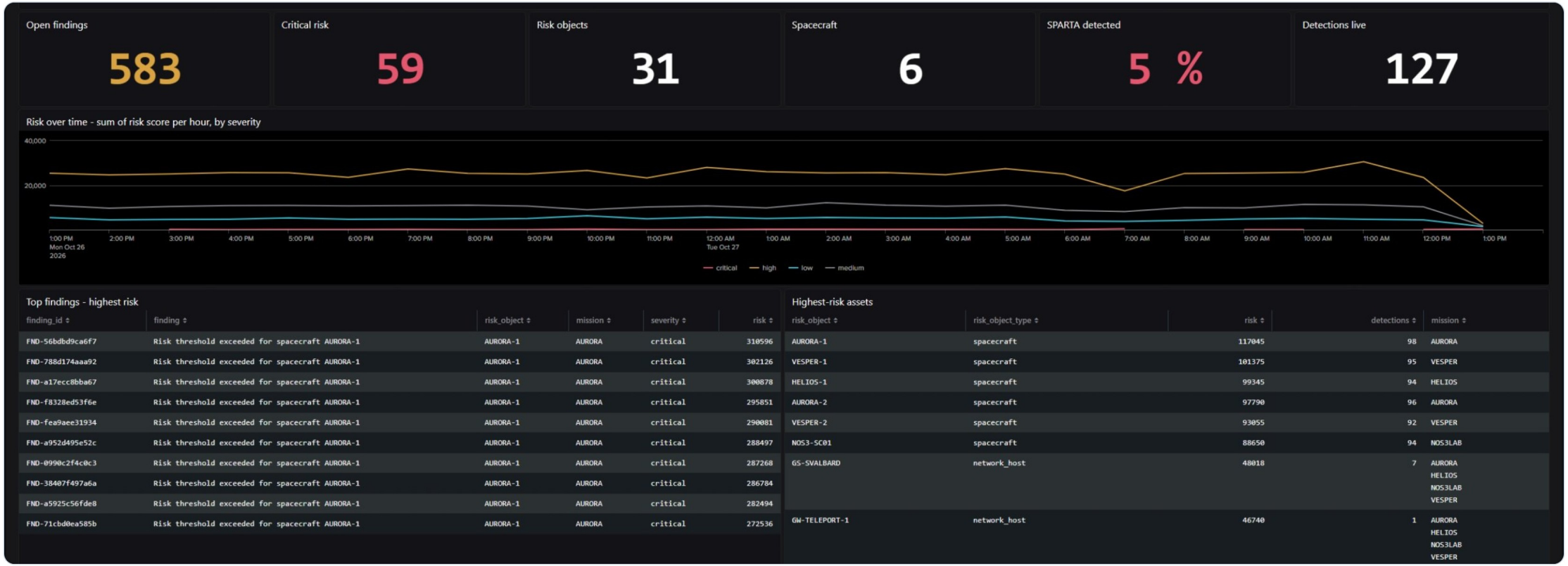
Job	Count	Schedule	Does
Detections	127	every 10 min	write scored risk events
RBA-001 promotion	1	every 10 min	promote high-fidelity hits
RBA-002, 005, 006	3	hourly	object, mission, cross-segment
RBA-003, 004	2	every 2 h and 4 h	7-day tactic and low-and-slow
Attack chains	6	hourly at :23	ordered multi-stage stories
Coverage and health	4	15 min to daily	SPARTA coverage, freshness
Demo generators	15	every 15 to 60 min	synthetic data, demo registries
Hunts	22	on demand	analyst-run searches
Setup and backfill	4	run once	demo missions, assets, history

FINDING YOUR WAY

The app menu, in the order of this tour

Menu	Consoles
Top level	Overview · Security Posture · Executive Summary
Operations	SOC Operations · Findings Review · Incident Review · Investigations · Mission Operations · Investigation Workbench · Attack Chains · Risk Analysis
Investigators	Spacecraft Investigator · Operator Investigator
Security Domains	Command Security · RF Security · GNSS / Navigation Security · Spacecraft Security · Ground Segment Security
Frameworks	SPARTA Coverage Matrix · Framework Coverage · Use Case Library · Content Metrics
Hunting and Engineering	Threat Hunting · Detection Engineering · Telemetry Health
Geospatial	Orbital View
Threat Intel and Response	Threat Intelligence
Configure	Content Management · Content Versioning · Asset & Identity · Command Catalog · Search
Audit	Incident Review Audit · Suppression Audit · Telemetry Health

Overview



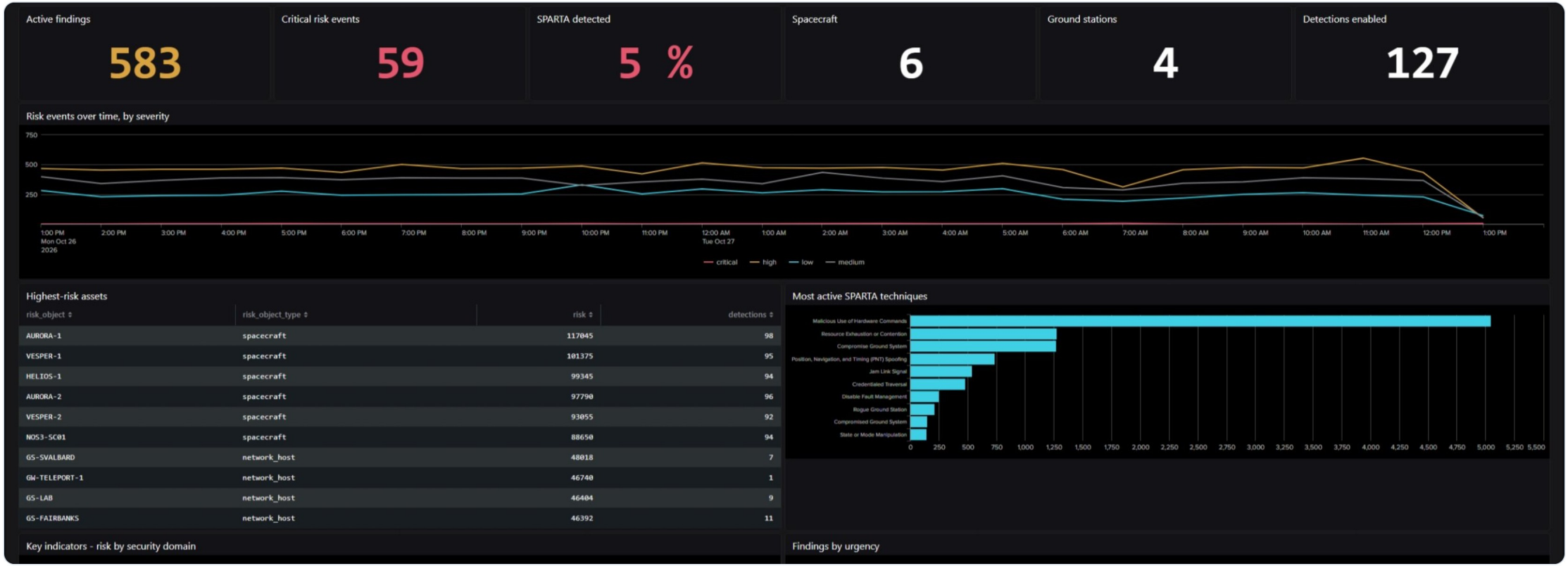
ON SCREEN

- Six KPIs: open findings, critical risk, risk objects, spacecraft, SPARTA coverage, live detections
- Risk per hour by severity, top findings, highest-risk assets
- Most active SPARTA techniques and 7-day attack chains

BEHIND IT

- Findings from satsec_findings, risk from satsec_risk
- Coverage from the SPARTA coverage KV, recomputed hourly
- Every tile and row drills into Incident Review or an investigator

Security Posture



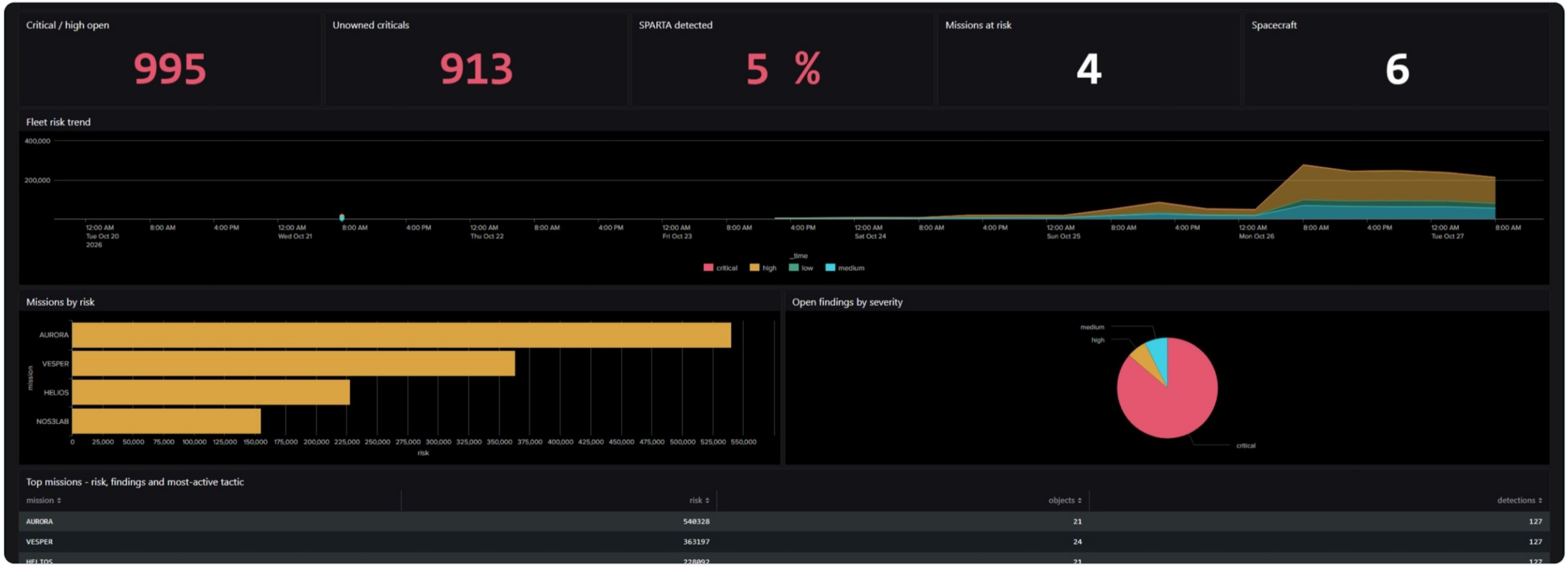
ON SCREEN

- KPIs incl. ground stations and detections enabled
- Risk by security domain and over time; findings by urgency
- Top threat sources and risk by mission

BEHIND IT

- Same risk and findings data as Overview, cut by domain
- Urgency = severity × asset criticality
- Threat sources: the threat objects detections record

Executive Summary



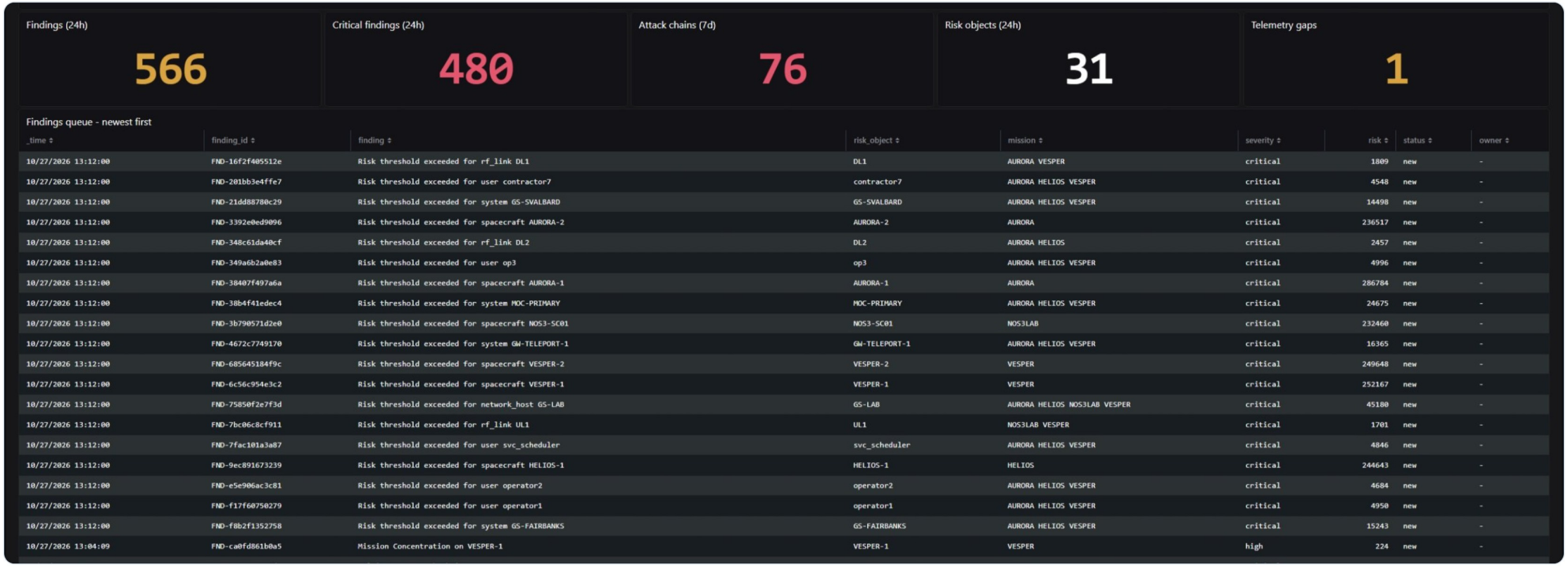
ON SCREEN

- Critical and high open, unowned criticals, coverage, missions at risk
- Fleet risk trend over the week, by severity
- Missions by risk, findings by severity, top missions

BEHIND IT

- “Unowned” means no owner in satsec_finding_reviews
- Defaults to the last 7 days for leadership reporting
- Top missions, with findings and the most active tactic

SOC Operations



_time	finding_id	finding	risk_object	mission	severity	risk	status	owner
10/27/2026 13:12:00	FND-16f2f405512e	Risk threshold exceeded for rf_link DL1	DL1	AURORA VESPER	critical	1809	new	-
10/27/2026 13:12:00	FND-201bb3e4ffe7	Risk threshold exceeded for user contractor7	contractor7	AURORA HELIOS VESPER	critical	4548	new	-
10/27/2026 13:12:00	FND-21dd88780c29	Risk threshold exceeded for system GS-SVALBARD	GS-SVALBARD	AURORA HELIOS VESPER	critical	14498	new	-
10/27/2026 13:12:00	FND-3392e0ed9096	Risk threshold exceeded for spacecraft AURORA-2	AURORA-2	AURORA	critical	236517	new	-
10/27/2026 13:12:00	FND-348c61da40cf	Risk threshold exceeded for rf_link DL2	DL2	AURORA HELIOS	critical	2457	new	-
10/27/2026 13:12:00	FND-349a6b2a0e83	Risk threshold exceeded for user op3	op3	AURORA HELIOS VESPER	critical	4996	new	-
10/27/2026 13:12:00	FND-38407f497a6a	Risk threshold exceeded for spacecraft AURORA-1	AURORA-1	AURORA	critical	286784	new	-
10/27/2026 13:12:00	FND-38b4f41edec4	Risk threshold exceeded for system MOC-PRIMARY	MOC-PRIMARY	AURORA HELIOS VESPER	critical	24675	new	-
10/27/2026 13:12:00	FND-3b790571d2e0	Risk threshold exceeded for spacecraft NOS3-SC01	NOS3-SC01	NOS3LAB	critical	232460	new	-
10/27/2026 13:12:00	FND-4672c7749170	Risk threshold exceeded for system GW-TELEPORT-1	GW-TELEPORT-1	AURORA HELIOS VESPER	critical	16365	new	-
10/27/2026 13:12:00	FND-685645184f9c	Risk threshold exceeded for spacecraft VESPER-2	VESPER-2	VESPER	critical	249648	new	-
10/27/2026 13:12:00	FND-6c56c954e3c2	Risk threshold exceeded for spacecraft VESPER-1	VESPER-1	VESPER	critical	252167	new	-
10/27/2026 13:12:00	FND-75850f2e7f3d	Risk threshold exceeded for network_host GS-LAB	GS-LAB	AURORA HELIOS NOS3LAB VESPER	critical	45180	new	-
10/27/2026 13:12:00	FND-7bc06c8cf911	Risk threshold exceeded for rf_link UL1	UL1	NOS3LAB VESPER	critical	1701	new	-
10/27/2026 13:12:00	FND-7fac101a3a87	Risk threshold exceeded for user svc_scheduler	svc_scheduler	AURORA HELIOS VESPER	critical	4846	new	-
10/27/2026 13:12:00	FND-9ec891673239	Risk threshold exceeded for spacecraft HELIOS-1	HELIOS-1	HELIOS	critical	244643	new	-
10/27/2026 13:12:00	FND-e5e906ac3c81	Risk threshold exceeded for user operator2	operator2	AURORA HELIOS VESPER	critical	4684	new	-
10/27/2026 13:12:00	FND-f17f60750279	Risk threshold exceeded for user operator1	operator1	AURORA HELIOS VESPER	critical	4950	new	-
10/27/2026 13:12:00	FND-f8b2f1352758	Risk threshold exceeded for system GS-FAIRBANKS	GS-FAIRBANKS	AURORA HELIOS VESPER	critical	15243	new	-
10/27/2026 13:04:09	FND-ca0fd861b0a5	Mission Concentration on VESPER-1	VESPER-1	VESPER	high	224	new	-

ON SCREEN

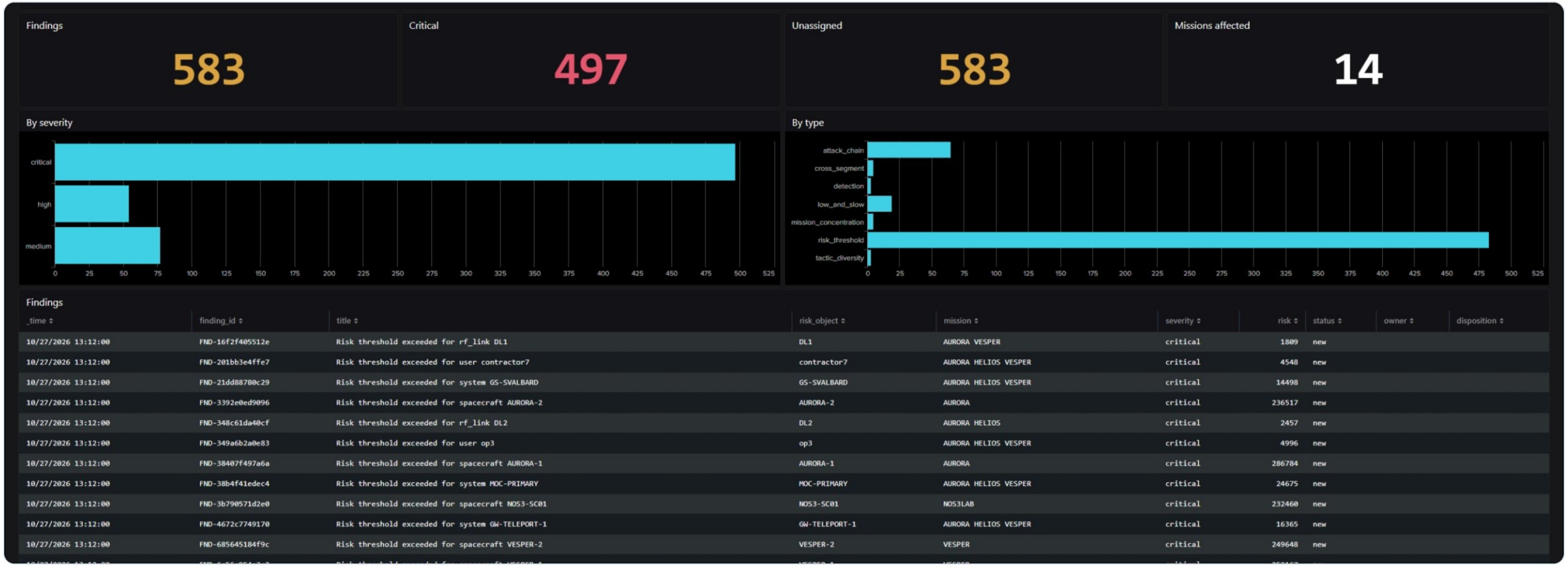
- Findings, criticals, chains, risk objects, telemetry gaps
- Newest-first findings queue and active attack chains
- Gaps, findings by rule, intel matches, suppressions

BEHIND IT

- Status comes from the review KV; untouched findings are New
- Gaps: sources silent for more than an hour (health summary)
- Intel matches are detection SAT-NET-011

Findings Review

This capture counted mission combinations; the true count is 4



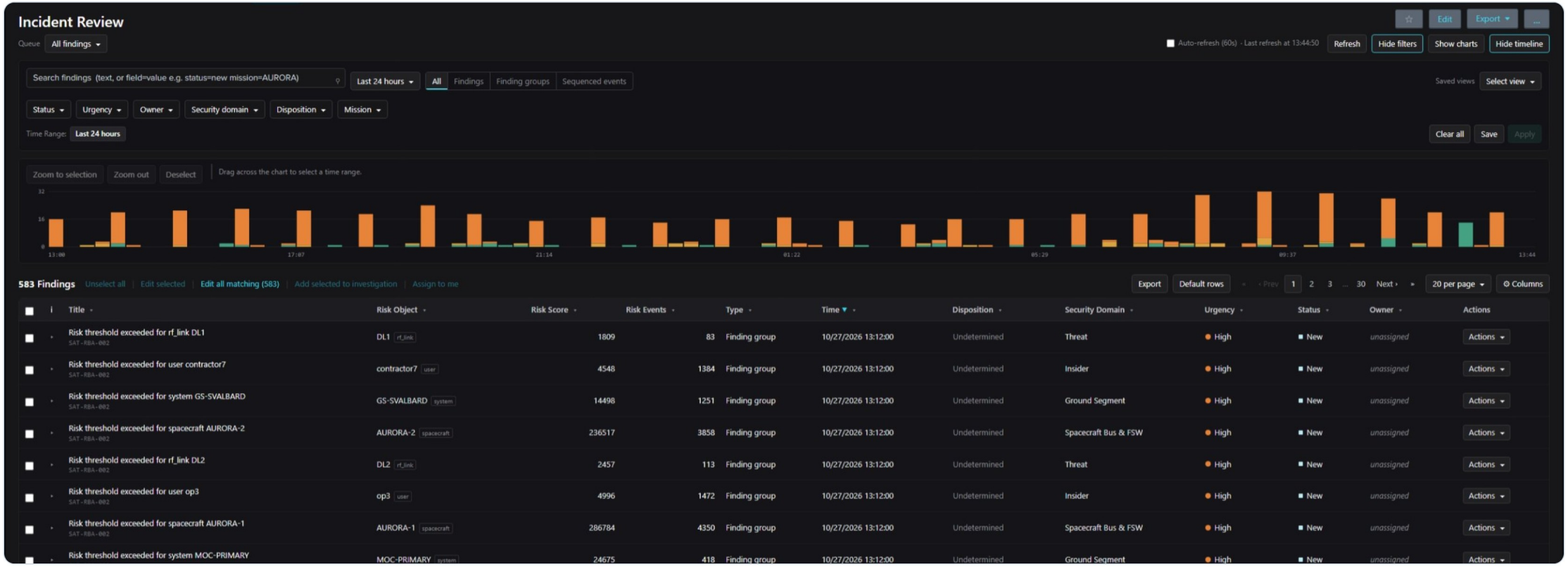
ON SCREEN

- Findings, critical, unassigned, missions affected
- Findings by severity and by type
- Full table with status, owner and disposition

BEHIND IT

- One row per finding from satsec_findings
- Review state joined from satsec_finding_reviews
- Click a row to open the finding in Incident Review

Incident Review



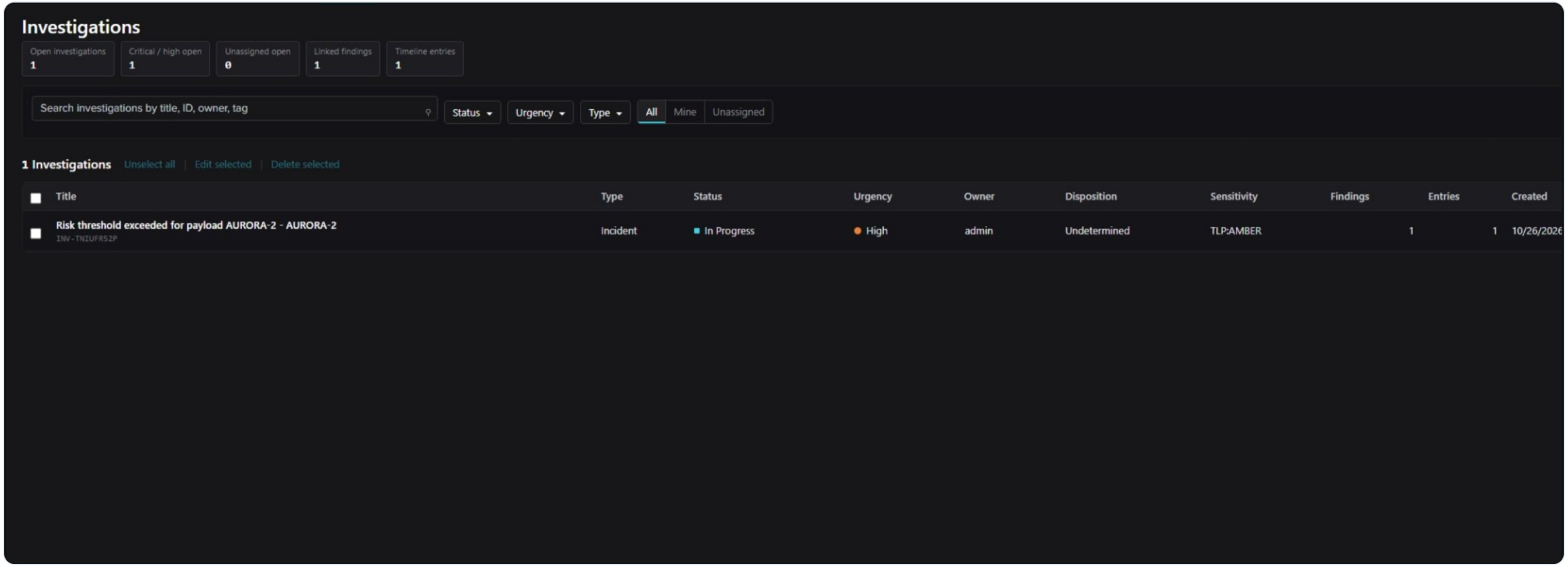
ON SCREEN

- Queue, search, time range, type and facet filters
- Urgency timeline: drag to select, zoom in and out
- Findings table with inline edits and bulk actions

BEHIND IT

- A JavaScript app: reads by search, writes by KV REST
- Urgency = severity × priority, with analyst override
- Every change appends to the review audit trail

Investigations



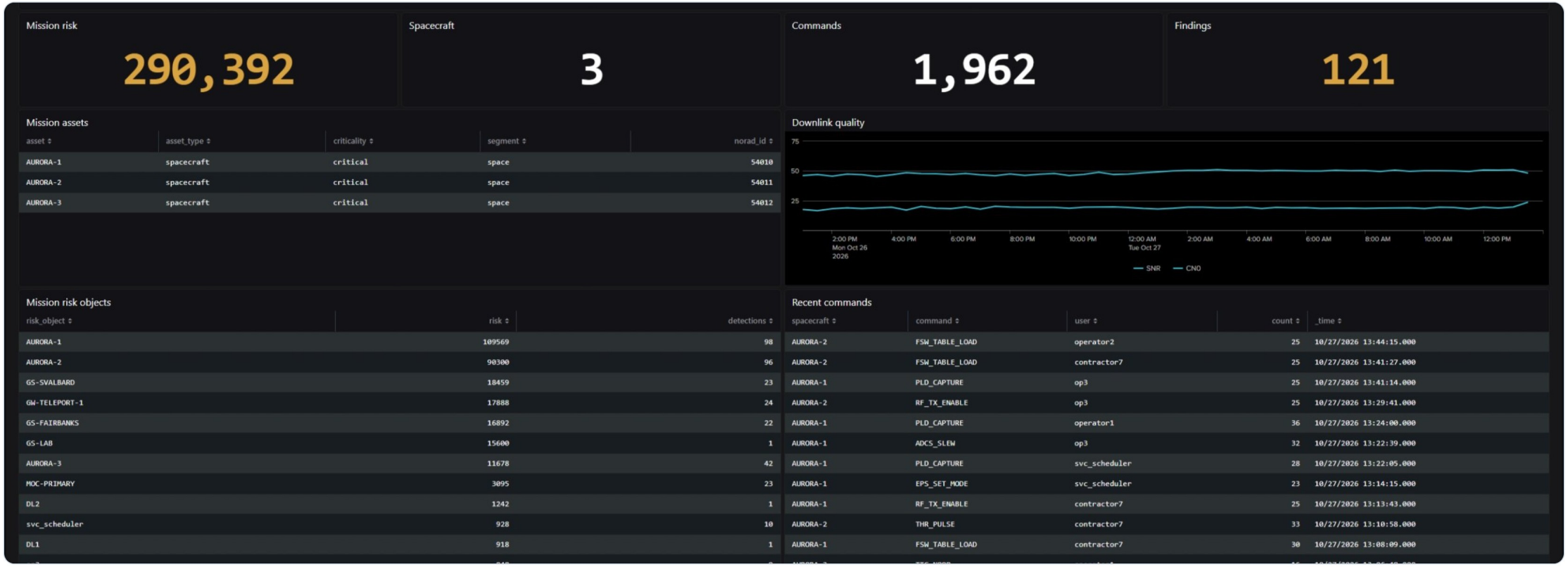
ON SCREEN

- Stats: open, critical/high, unassigned, findings, entries
- Search, status, urgency and type facets; All, Mine, Unassigned
- Click any row to open the investigation workbench

BEHIND IT

- satsec_investigations and _notes, written over KV REST
- Workbench: Overview, Timeline, Findings, Workbench, History
- Linking findings never overwrites their review state

Mission Operations



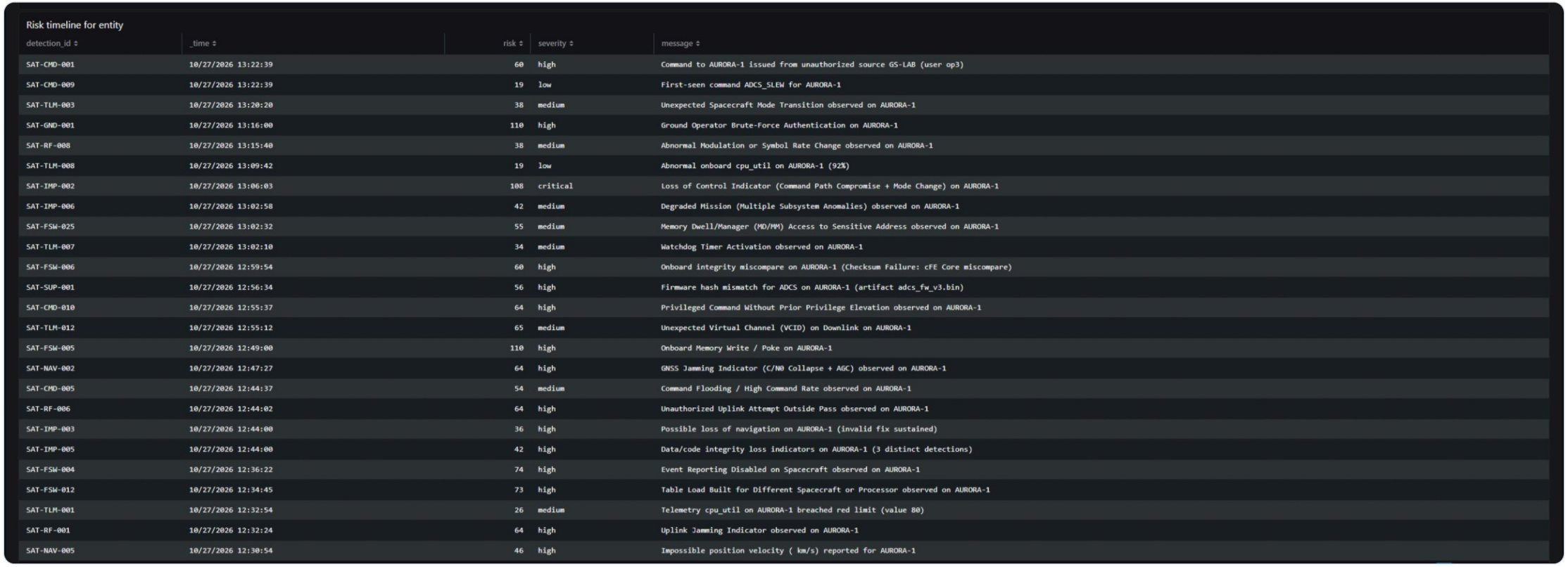
ON SCREEN

- Mission risk, spacecraft, commands and findings
- Mission assets and downlink quality (SNR, C/N0)
- Mission risk objects and recent commands

BEHIND IT

- Mission list from the mission registry (KV)
- Downlink from satsec_rf, commands from satsec_command
- Click a spacecraft or operator to open its investigator

Investigation Workbench



Risk timeline for entity

detection_id	_time	risk	severity	message
SAT-CMD-001	10/27/2026 13:22:39	60	high	Command to AURORA-1 issued from unauthorized source GS-LAB (user op3)
SAT-CMD-009	10/27/2026 13:22:39	19	low	First-seen command ADCS_SLEW for AURORA-1
SAT-TLM-003	10/27/2026 13:20:20	38	medium	Unexpected Spacecraft Mode Transition observed on AURORA-1
SAT-GND-001	10/27/2026 13:16:00	110	high	Ground Operator Brute-Force Authentication on AURORA-1
SAT-RF-008	10/27/2026 13:15:40	38	medium	Abnormal Modulation or Symbol Rate Change observed on AURORA-1
SAT-TLM-008	10/27/2026 13:09:42	19	low	Abnormal onboard cpu_util on AURORA-1 (92%)
SAT-IMP-002	10/27/2026 13:06:03	108	critical	Loss of Control Indicator (Command Path Compromise + Mode Change) on AURORA-1
SAT-IMP-006	10/27/2026 13:02:58	42	medium	Degraded Mission (Multiple Subsystem Anomalies) observed on AURORA-1
SAT-FSW-025	10/27/2026 13:02:32	55	medium	Memory Dwell/Manager (MD/MM) Access to Sensitive Address observed on AURORA-1
SAT-TLM-007	10/27/2026 13:02:10	34	medium	Watchdog Timer Activation observed on AURORA-1
SAT-FSW-006	10/27/2026 12:59:54	60	high	Onboard integrity miscompare on AURORA-1 (Checksum Failure: cFE Core miscompare)
SAT-SUP-001	10/27/2026 12:56:34	56	high	Firmware hash mismatch for ADCS on AURORA-1 (artifact adcs_fw_v3.bin)
SAT-CMD-010	10/27/2026 12:55:37	64	high	Privileged Command Without Prior Privilege Elevation observed on AURORA-1
SAT-TLM-012	10/27/2026 12:55:12	65	medium	Unexpected Virtual Channel (VCID) on Downlink on AURORA-1
SAT-FSW-005	10/27/2026 12:49:00	110	high	Onboard Memory Write / Poke on AURORA-1
SAT-NAV-002	10/27/2026 12:47:27	64	high	GNSS Jamming Indicator (C/N0 Collapse + AGC) observed on AURORA-1
SAT-CMD-005	10/27/2026 12:44:37	54	medium	Command Flooding / High Command Rate observed on AURORA-1
SAT-RF-006	10/27/2026 12:44:02	64	high	Unauthorized Uplink Attempt Outside Pass observed on AURORA-1
SAT-IMP-003	10/27/2026 12:44:00	36	high	Possible loss of navigation on AURORA-1 (invalid fix sustained)
SAT-IMP-005	10/27/2026 12:44:00	42	high	Data/code integrity loss indicators on AURORA-1 (3 distinct detections)
SAT-FSW-004	10/27/2026 12:36:22	74	high	Event Reporting Disabled on Spacecraft observed on AURORA-1
SAT-FSW-012	10/27/2026 12:34:45	73	high	Table Load Built for Different Spacecraft or Processor observed on AURORA-1
SAT-TLM-001	10/27/2026 12:32:54	26	medium	Telemetry cpu_util on AURORA-1 breached red limit (value 80)
SAT-RF-001	10/27/2026 12:32:24	64	high	Uplink Jamming Indicator observed on AURORA-1
SAT-NAV-005	10/27/2026 12:30:54	46	high	Impossible position velocity (km/s) reported for AURORA-1

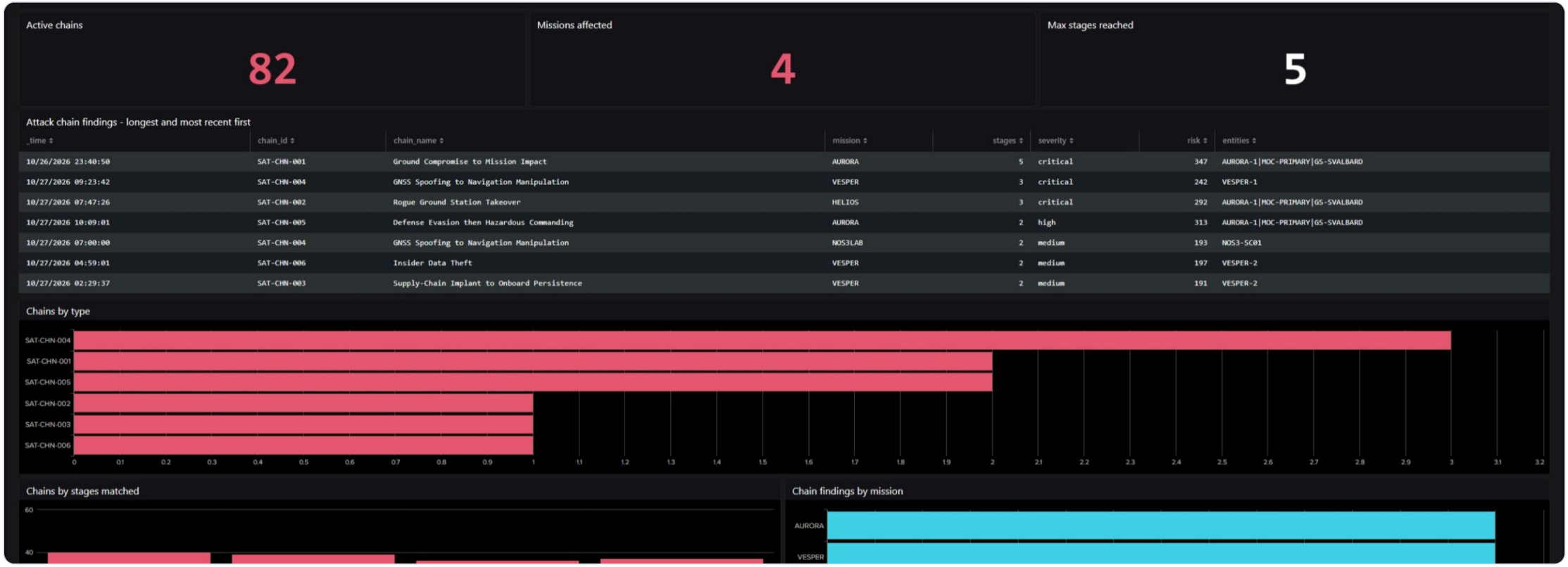
ON SCREEN

- One entity box: object, IP, user or spacecraft
- Risk timeline: every detection that touched it
- Its command activity and ground access

BEHIND IT

- Matches risk_object or threat_object in satsec_risk
- Command and ground panels match user, spacecraft or source
- Rows pivot into the Spacecraft or Operator Investigator

Attack Chains



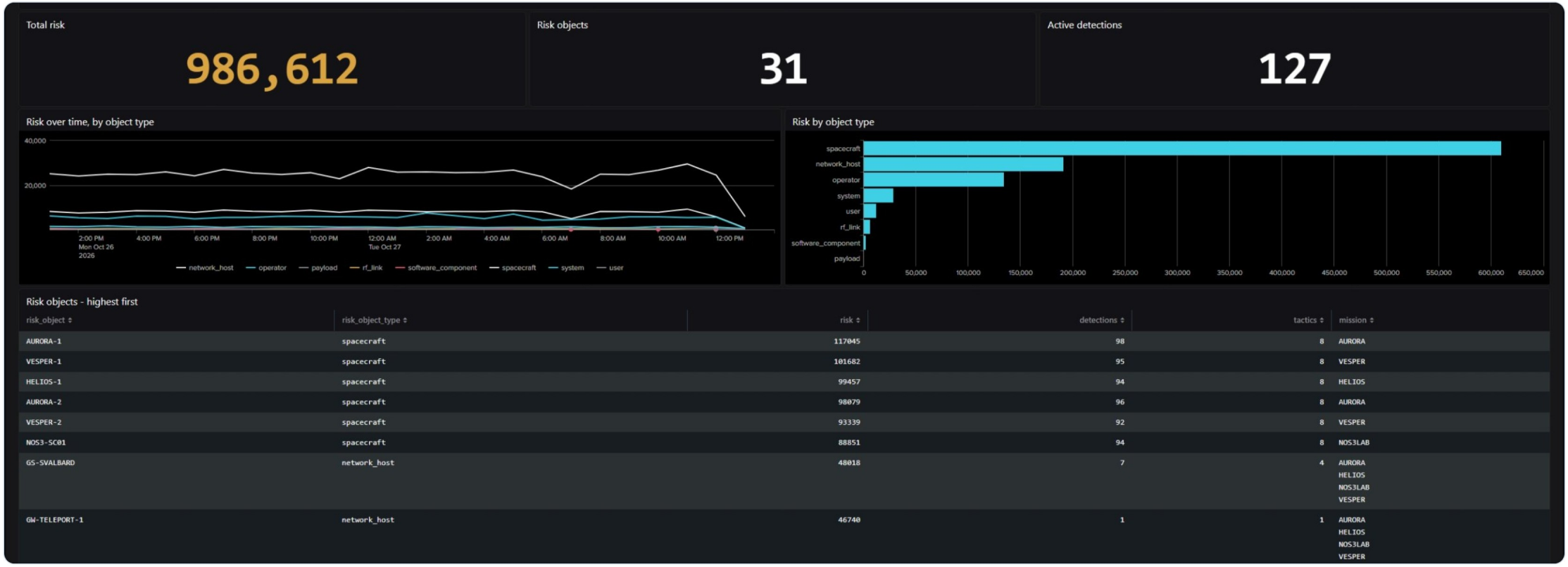
ON SCREEN

- Active chains, missions affected, max stages reached
- Chain findings: mission, stages, severity, risk, entities
- Chains by type, by stages matched and by mission

BEHIND IT

- Six correlations map risk events to chain stages
- A chain fires when stages occur in order on a mission
- Findings land in Incident Review as Sequenced events

Risk Analysis



ON SCREEN

- Total risk, risk objects, active detections
- Risk over time and by object type
- Risk objects; risk by tactic; top detections; RBA rules

BEHIND IT

- Every row is a scored risk event in satsec_risk
- Scores weighted by asset, mission and mission phase
- Findings by RBA rule shows which rules are firing

Spacecraft Investigator

ON SCREEN

- Risk, findings, commands, red limits, auth failures
- Risk by SPARTA tactic; commanding by auth state; RF
- Findings, commands, limit breaches, ground contacts

BEHIND IT

- The ES Asset Investigator, keyed on the spacecraft
- Joins risk, findings, command, telemetry, RF and contacts
- Click an operator or a finding to keep pivoting

Operator Investigator

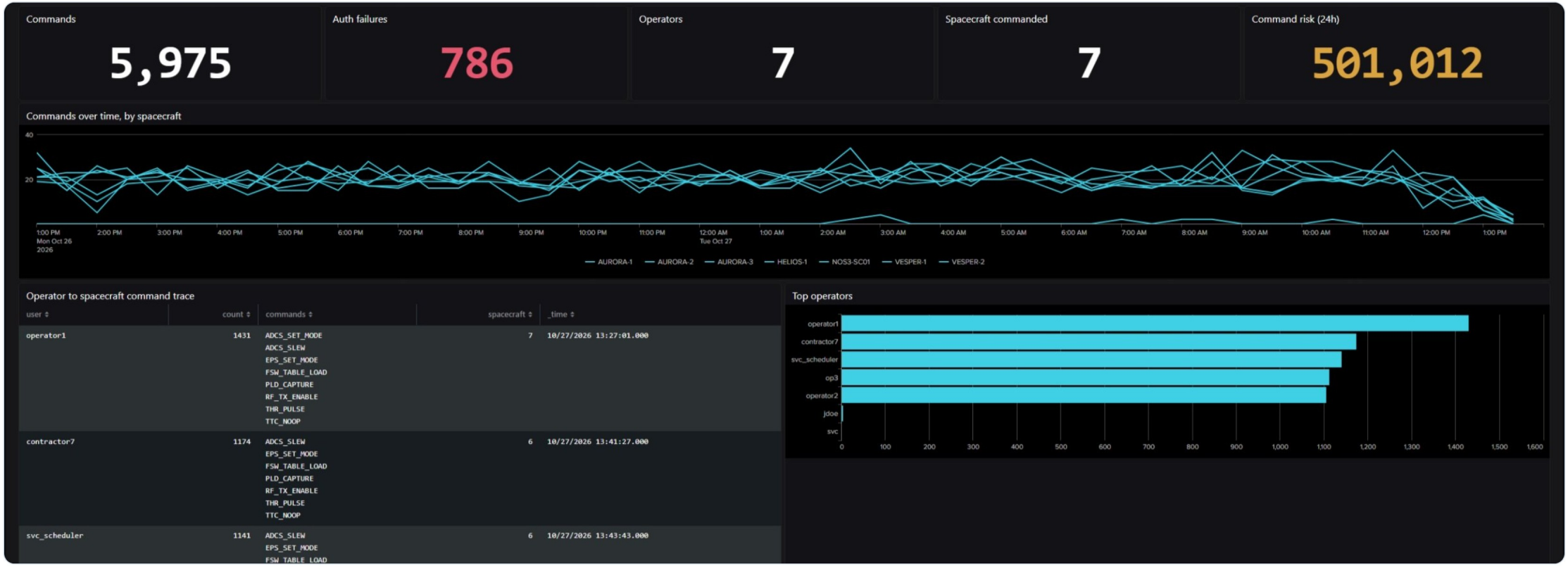
ON SCREEN

- Risk, commands, spacecraft, auth failures, findings
- Risk by tactic, commanding by spacecraft, logins
- Commands issued and findings attributed to them

BEHIND IT

- The ES Identity Investigator, keyed on the operator
- Operators come from commanding, logins and risk
- Riskiest operator first, selected automatically

Command Security



ON SCREEN

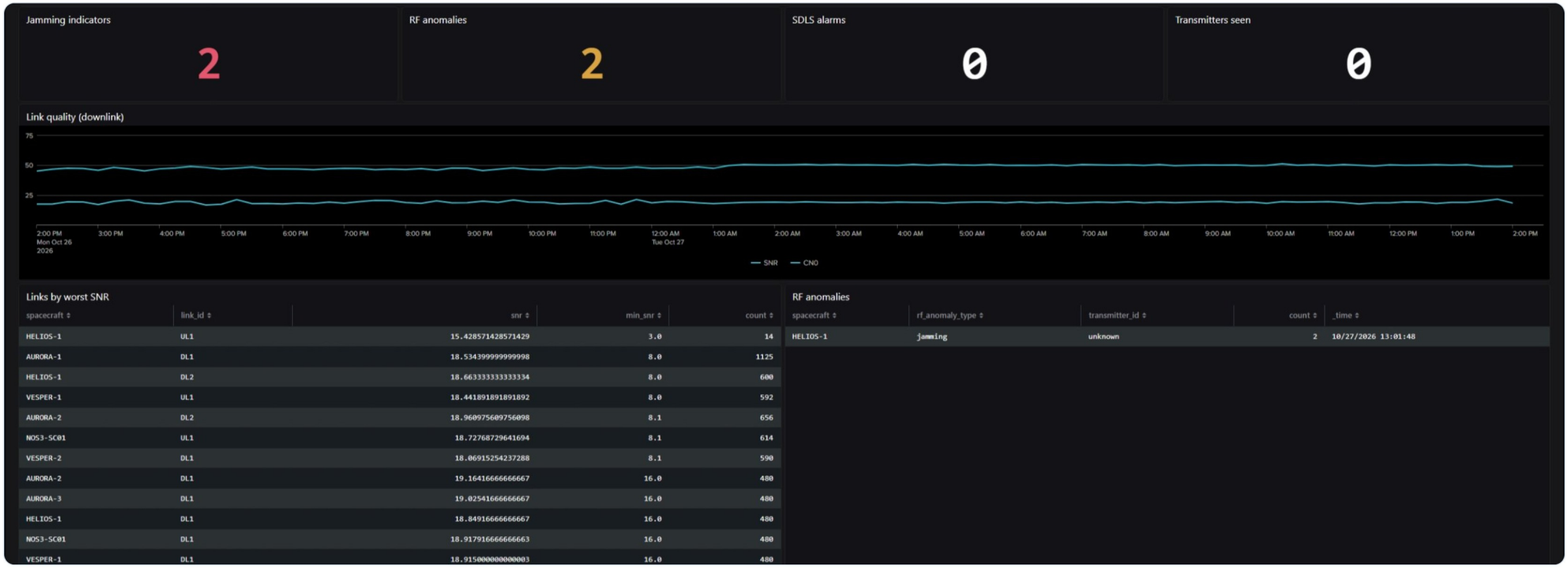
- Commands, auth failures, operators, spacecraft, command risk
- Commands over time by spacecraft
- Operator-to-spacecraft trace, top operators, risk table

BEHIND IT

- Telecommands in satsec_command: user, source, auth state
- SAT-CMD-001 to -010 score off-window, unauthorized, hazardous
- Hazard levels come from the Command Catalog

RF Security

Transmitter and SDLS alarm counts were still filling in at capture time



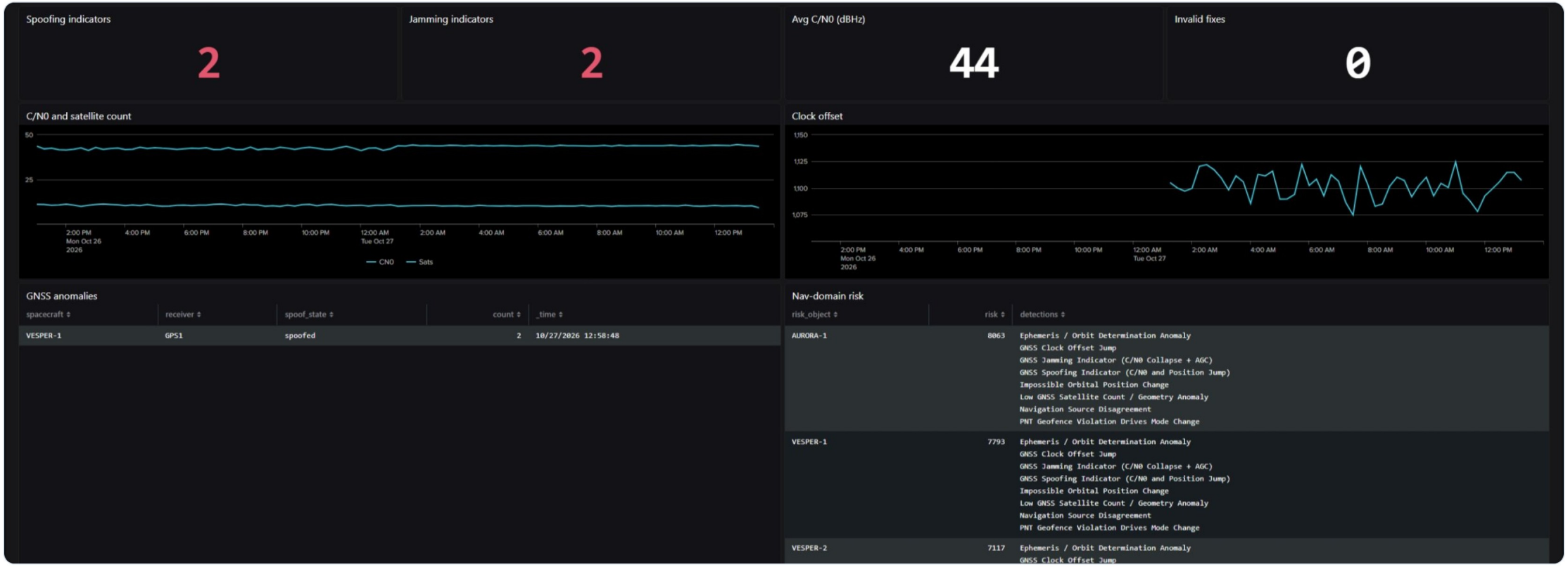
ON SCREEN

- Jamming, RF anomalies, SDLS alarms, transmitters seen
- Downlink SNR and C/N0 over time
- Links by worst SNR, anomalies, RF and link risk

BEHIND IT

- Link metrics and SDLS events in satsec_rf
- SAT-RF-* and SAT-LNK-*: drops, jamming, replay, MAC failures
- SDR bridge and CCSDS decoder feed real links

GNSS / Navigation Security



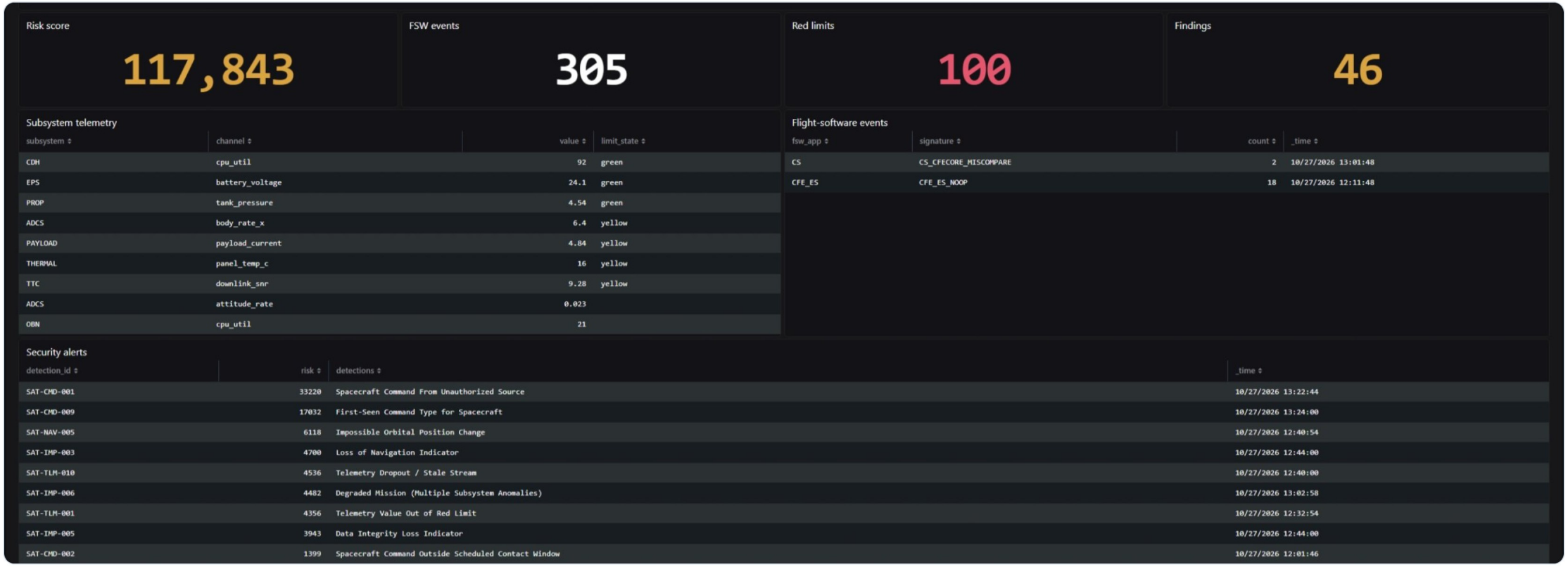
ON SCREEN

- Spoofing and jamming indicators, C/N0, invalid fixes
- C/N0 with satellite count; receiver clock offset
- GNSS anomalies and nav-domain risk per spacecraft

BEHIND IT

- Receiver telemetry in satsec_nav (NMEA adapter)
- SAT-NAV-*: spoofing, jamming, impossible position, clock jumps
- Clock jumps often appear before position spoofing

Spacecraft Security



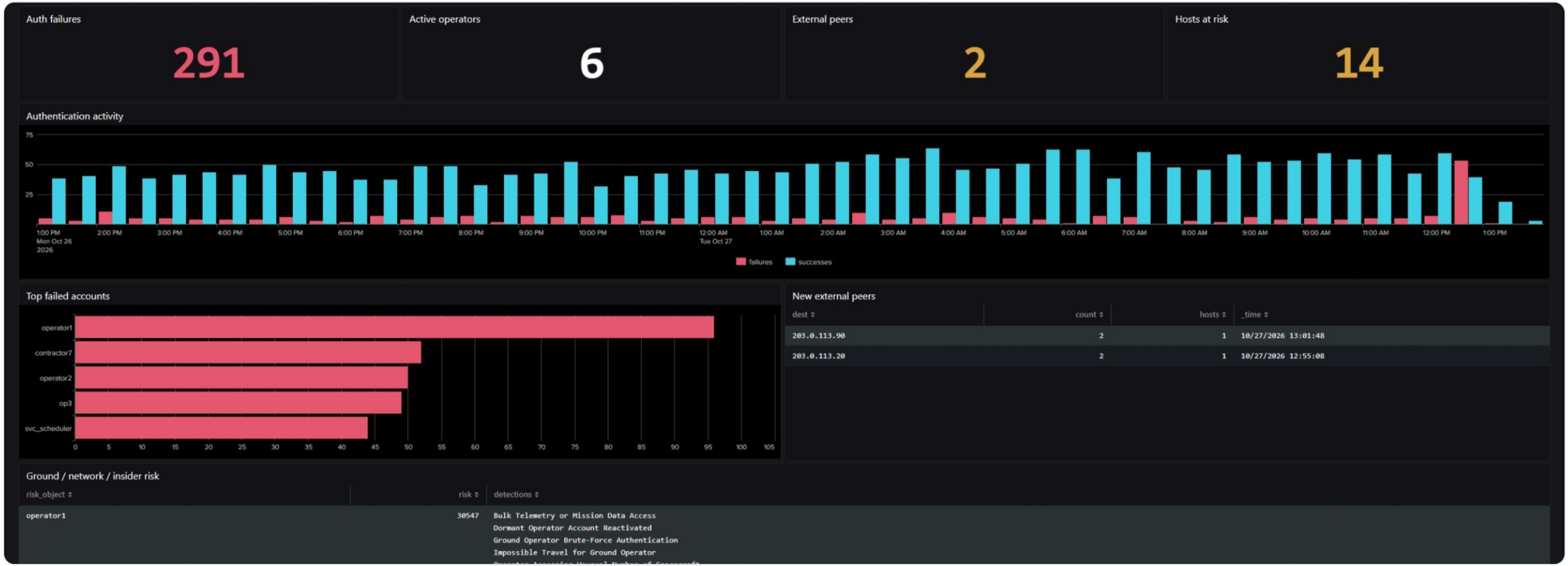
ON SCREEN

- Risk score, FSW events, red limits, findings
- Subsystem telemetry with limit state
- Flight-software events and alerts by detection

BEHIND IT

- Telemetry and cFS/cFE events in satsec_spacecraft
- SAT-FSW-* and SAT-TLM-* watch integrity and limits
- NOS3 simulator events land here too

Ground Segment Security



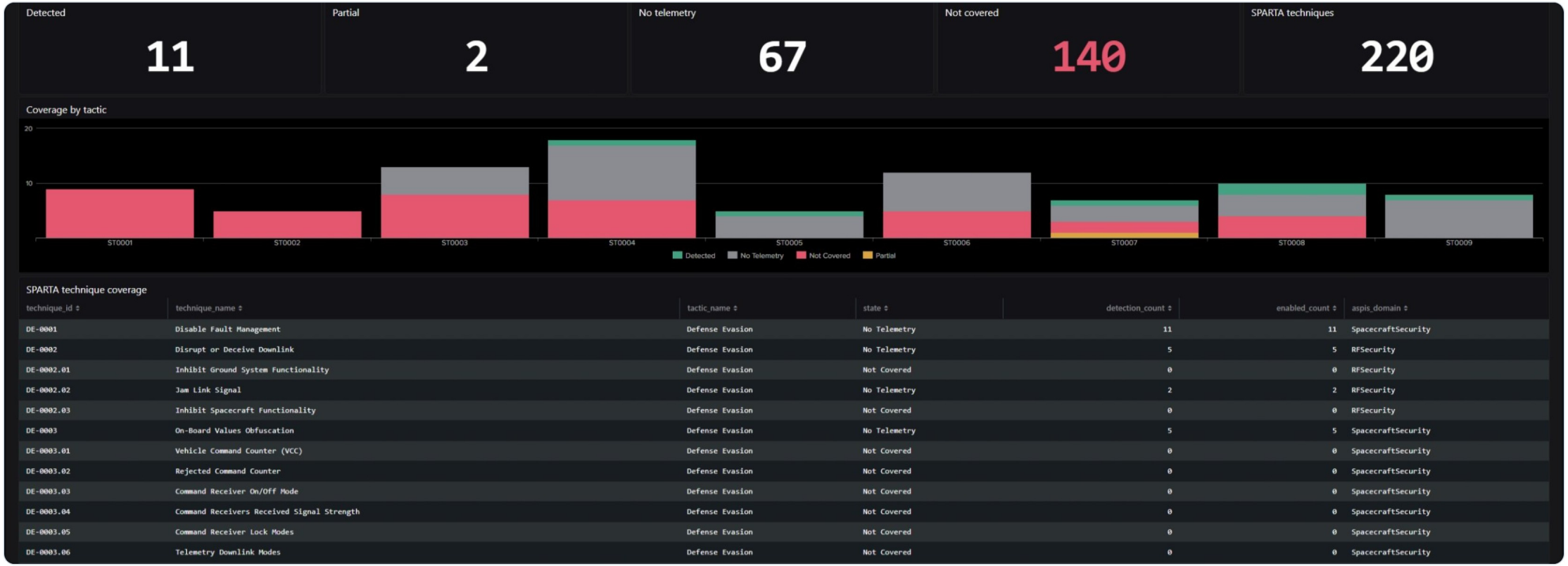
ON SCREEN

- Auth failures, operators, external peers, hosts at risk
- Login successes and failures over time
- Failed accounts, new external peers, ground risk

BEHIND IT

- Logins and network flows in satsec_ground
- SAT-GND-*, SAT-NET-*, SAT-INS-*: abuse, beaconing, insiders
- CIM-tagged, so standard authentication searches work

SPARTA Coverage Matrix



SPARTA technique coverage

technique_id	technique_name	tactic_name	state	detection_count	enabled_count	aspis_domain
DE-0001	Disable Fault Management	Defense Evasion	No Telemetry	11	11	SpacecraftSecurity
DE-0002	Disrupt or Deceive Downlink	Defense Evasion	No Telemetry	5	5	RFSecurity
DE-0002.01	Inhibit Ground System Functionality	Defense Evasion	Not Covered	0	0	RFSecurity
DE-0002.02	Jam Link Signal	Defense Evasion	No Telemetry	2	2	RFSecurity
DE-0002.03	Inhibit Spacecraft Functionality	Defense Evasion	Not Covered	0	0	RFSecurity
DE-0003	On-Board Values Obfuscation	Defense Evasion	No Telemetry	5	5	SpacecraftSecurity
DE-0003.01	Vehicle Command Counter (VCC)	Defense Evasion	Not Covered	0	0	SpacecraftSecurity
DE-0003.02	Rejected Command Counter	Defense Evasion	Not Covered	0	0	SpacecraftSecurity
DE-0003.03	Command Receiver On/Off Mode	Defense Evasion	Not Covered	0	0	SpacecraftSecurity
DE-0003.04	Command Receivers Received Signal Strength	Defense Evasion	Not Covered	0	0	SpacecraftSecurity
DE-0003.05	Command Receiver Lock Modes	Defense Evasion	Not Covered	0	0	SpacecraftSecurity
DE-0003.06	Telemetry Downlink Modes	Defense Evasion	Not Covered	0	0	SpacecraftSecurity

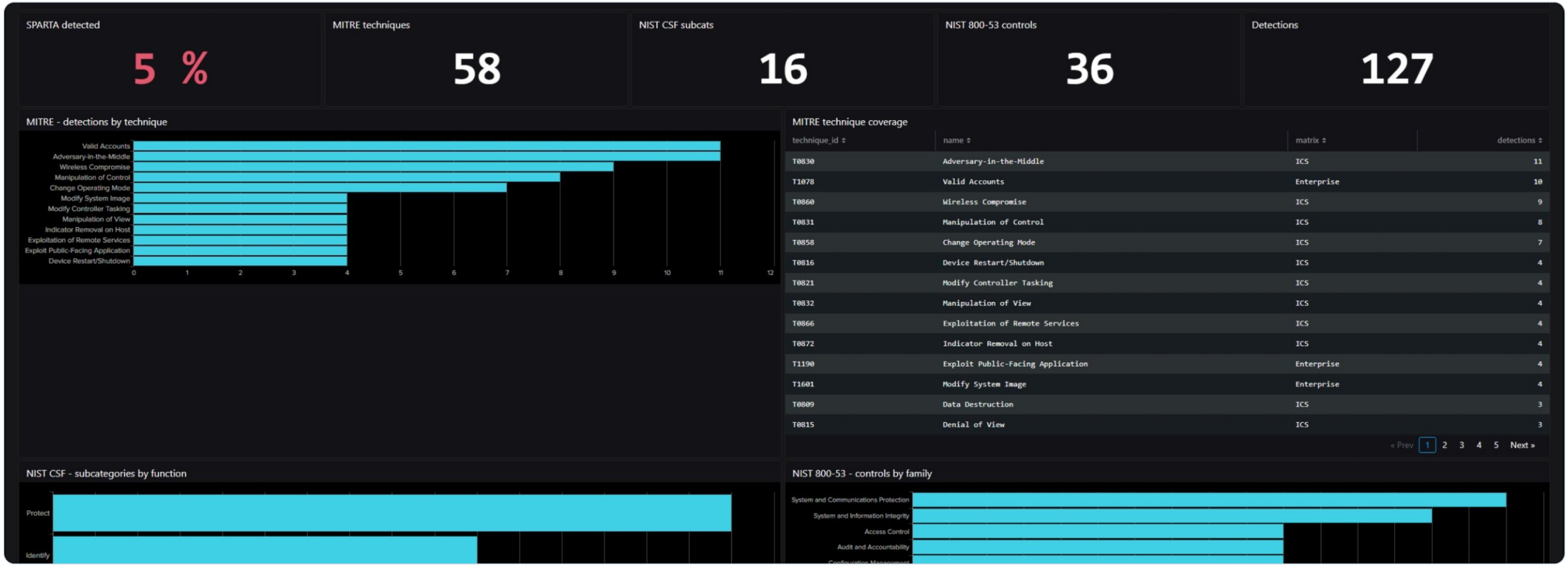
ON SCREEN

- Techniques detected, partial, no telemetry, not covered
- Coverage state per SPARTA tactic
- Per-technique state, detection counts, ASPIS domain

BEHIND IT

- Coverage KV recomputed hourly from content and freshness
- Detected: a full detection plus fresh required telemetry
- No telemetry: detections exist, data is not arriving

Framework Coverage



ON SCREEN

- SPARTA %, ATT&CK techniques, CSF subcategories, controls
- ATT&CK detections by technique and coverage table
- NIST CSF by function; 800-53 by control family

BEHIND IT

- Mappings ship in each detection’s YAML and the catalog
- ATT&CK, CSF and 800-53 lookups name every ID
- Content view only: it needs no telemetry

Use Case Library

ON SCREEN

- Detections, techniques covered, stories, production count
- Detections by SPARTA tactic and by domain
- Attack-chain stories and the full detection catalogue

BEHIND IT

- Catalog lookup built from the YAML content at build time
- Stories come from the chain-stage lookup
- Each story maps chain stages to their detections

Content Metrics

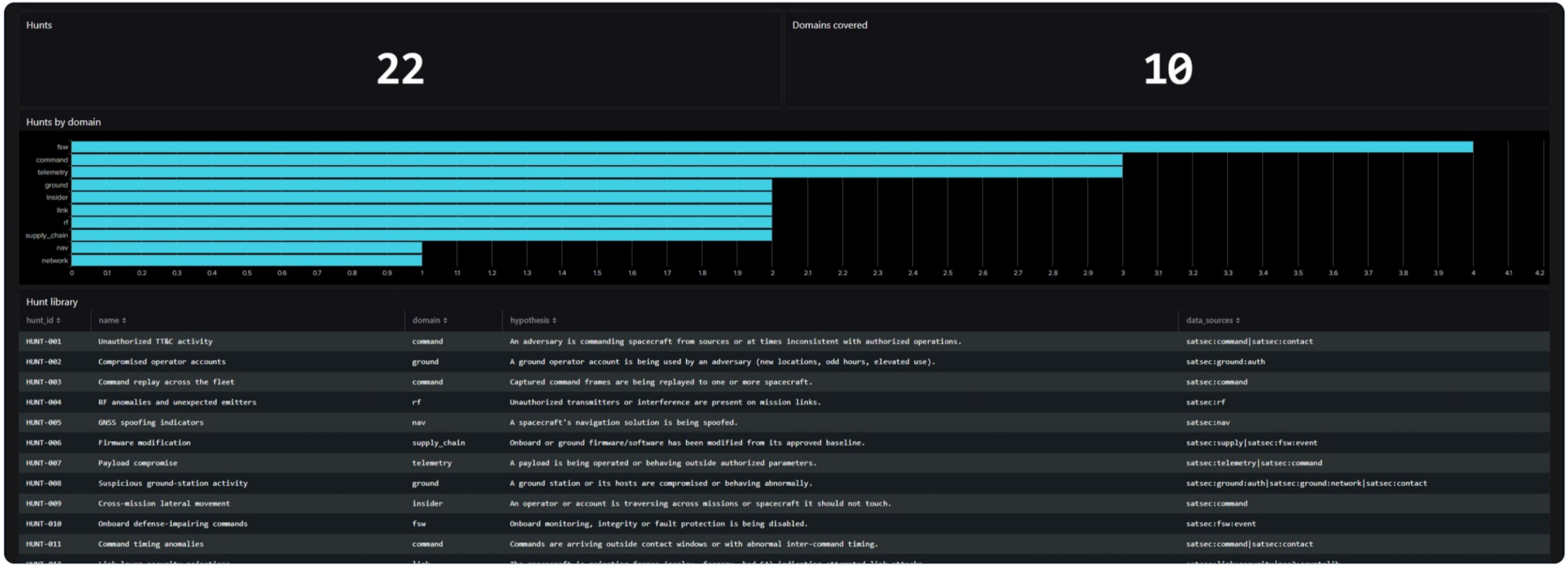
ON SCREEN

- SPARTA techniques, techniques with detections
- Detections and SPARTA countermeasures
- Detections by tactic and by data source

BEHIND IT

- The SPARTA knowledge base ships as lookups
- A detection-to-technique lookup links content to SPARTA
- A daily snapshot job records content metrics

Threat Hunting



hunt_id	name	domain	hypothesis	data_sources
HUNT-001	Unauthorized TT&C activity	command	An adversary is commanding spacecraft from sources or at times inconsistent with authorized operations.	satsec:command|satsec:contact
HUNT-002	Compromised operator accounts	ground	A ground operator account is being used by an adversary (new locations, odd hours, elevated use).	satsec:ground:auth
HUNT-003	Command replay across the fleet	command	Captured command frames are being replayed to one or more spacecraft.	satsec:command
HUNT-004	RF anomalies and unexpected emitters	rf	Unauthorized transmitters or interference are present on mission links.	satsec:rf
HUNT-005	GNSS spoofing indicators	nav	A spacecraft's navigation solution is being spoofed.	satsec:nav
HUNT-006	Firmware modification	supply_chain	Onboard or ground firmware/software has been modified from its approved baseline.	satsec:supply|satsec:fsw:event
HUNT-007	Payload compromise	telemetry	A payload is being operated or behaving outside authorized parameters.	satsec:telemetry|satsec:command
HUNT-008	Suspicious ground-station activity	ground	A ground station or its hosts are compromised or behaving abnormally.	satsec:ground:auth|satsec:ground:network|satsec:contact
HUNT-009	Cross-mission lateral movement	insider	An operator or account is traversing across missions or spacecraft it should not touch.	satsec:command
HUNT-010	Onboard defense-impairing commands	fsw	Onboard monitoring, integrity or fault protection is being disabled.	satsec:fsw:event
HUNT-011	Command timing anomalies	command	Commands are arriving outside contact windows or with abnormal inter-command timing.	satsec:command|satsec:contact

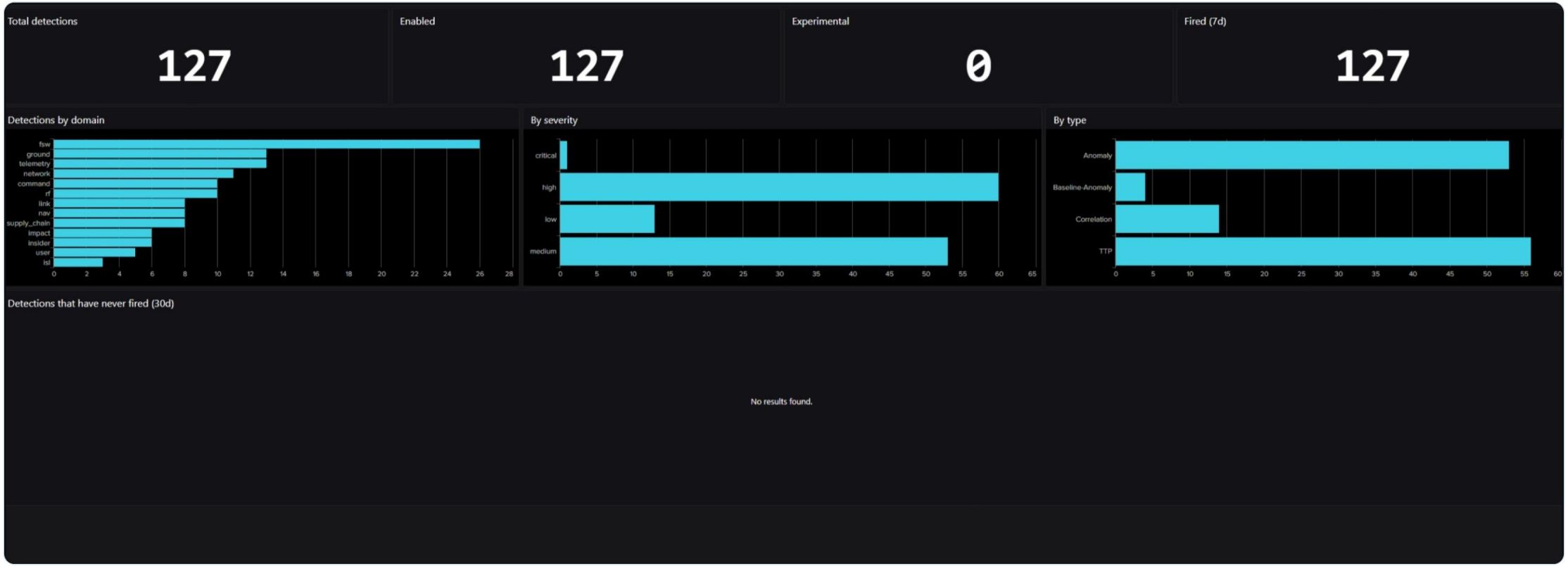
ON SCREEN

- 22 hunts across 10 domains
- Hunts by domain
- Library: hypothesis and the data each hunt needs

BEHIND IT

- Each hunt is a saved search “ASPIS Hunt - id - name”
- Run on demand from Search; never scheduled
- Click a hunt to open and run it

Detection Engineering



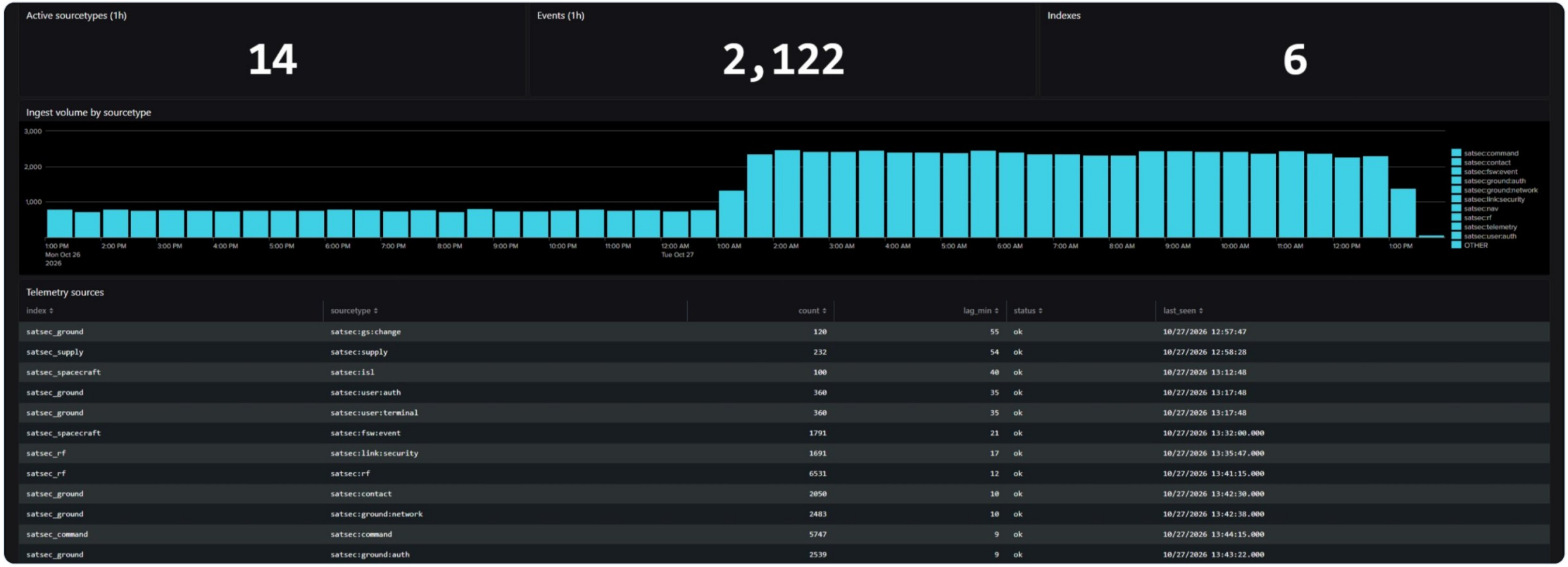
ON SCREEN

- Total, enabled, experimental, fired in 7 days
- Detections by domain, severity and type
- Detections that have never fired in 30 days

BEHIND IT

- Catalog lookup joined with activity in satsec_risk
- Never-fired list exposes dead content or missing data
- Empty here because the demo exercises all 127

Telemetry Health



index	sourcetype	count	lag_min	status	last_seen
satsec_ground	satsec:gs:change	120	55	ok	10/27/2026 12:57:47
satsec_supply	satsec:supply	232	54	ok	10/27/2026 12:58:28
satsec_spacecraft	satsec:isl	100	40	ok	10/27/2026 13:12:48
satsec_ground	satsec:user:auth	360	35	ok	10/27/2026 13:17:48
satsec_ground	satsec:user:terminal	360	35	ok	10/27/2026 13:17:48
satsec_spacecraft	satsec:fsw:event	1791	21	ok	10/27/2026 13:32:00.000
satsec_rf	satsec:link:security	1691	17	ok	10/27/2026 13:35:47.000
satsec_rf	satsec:rf	6531	12	ok	10/27/2026 13:41:15.000
satsec_ground	satsec:contact	2050	10	ok	10/27/2026 13:42:30.000
satsec_ground	satsec:ground:network	2483	10	ok	10/27/2026 13:42:38.000
satsec_command	satsec:command	5747	9	ok	10/27/2026 13:44:15.000
satsec_ground	satsec:ground:auth	2539	9	ok	10/27/2026 13:43:22.000

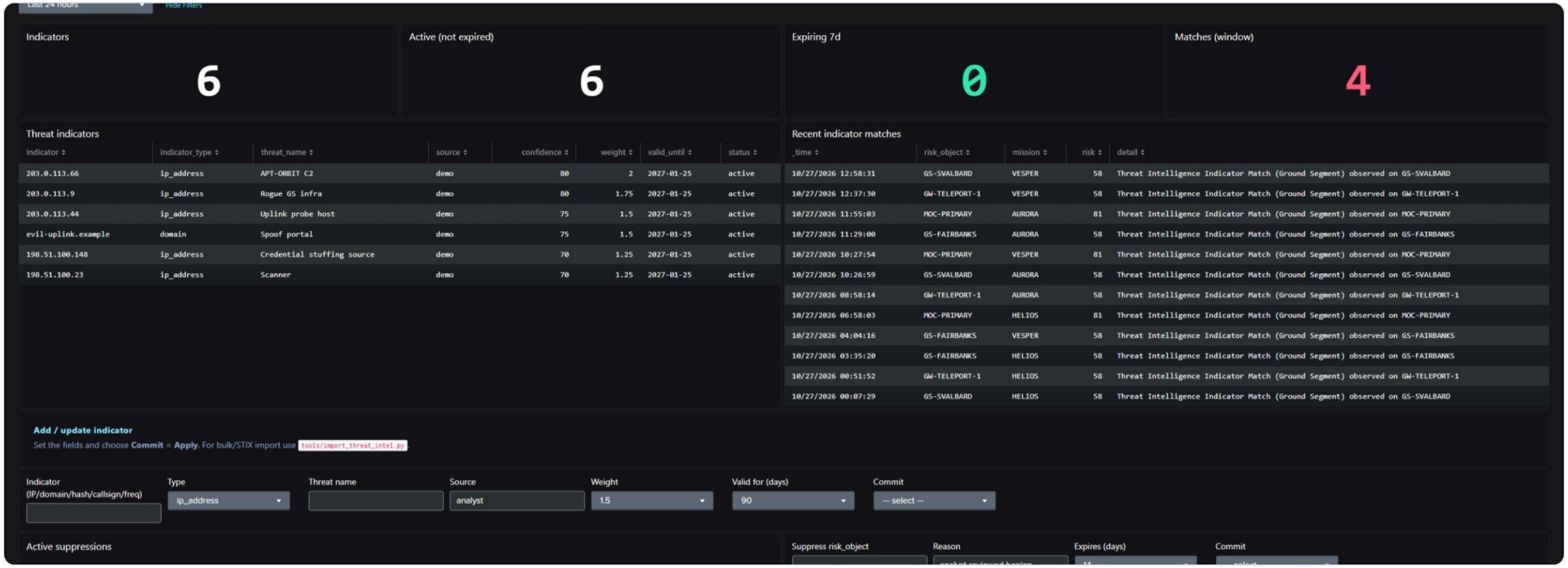
ON SCREEN

- Active sourcetypes, events and indexes (last hour)
- Ingest volume by sourcetype
- Every source: count, lag, OK or STALE, last seen

BEHIND IT

- Reads all six telemetry indexes through one macro
- Freshness also drives SPARTA “No telemetry” states
- Listed under both Engineering and Audit menus

Threat Intelligence



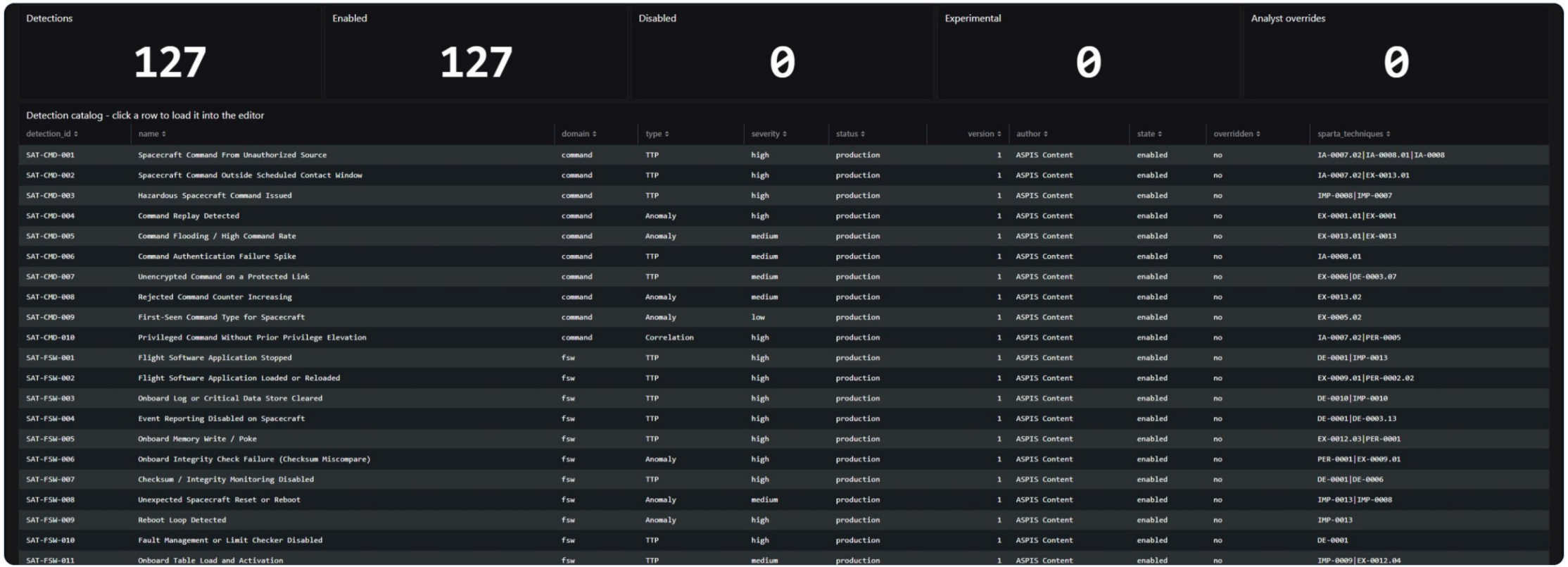
ON SCREEN

- Indicators, active, expiring, matches in the window
- Indicator list and recent matches
- Indicator and suppression forms; SBOM and vulnerabilities

BEHIND IT

- Indicators in satsec_threat_intel, matched by SAT-NET-011
- A match also multiplies a detection’s risk by 1.5
- Bulk CSV or STIX import: tools/import_threat_intel.py

Content Management



Detections	Enabled	Disabled	Experimental	Analyst overrides
127	127	0	0	0

Detection catalog - click a row to load it into the editor

detection_id	name	domain	type	severity	status	version	author	state	overridden	sparta_techniques
SAT-CMD-001	Spacecraft Command From Unauthorized Source	command	TTP	high	production	1	ASPIS Content	enabled	no	IA-0007.02|IA-0008.01|IA-0008
SAT-CMD-002	Spacecraft Command Outside Scheduled Contact Window	command	TTP	high	production	1	ASPIS Content	enabled	no	IA-0007.02|EX-0013.01
SAT-CMD-003	Hazardous Spacecraft Command Issued	command	TTP	high	production	1	ASPIS Content	enabled	no	IMP-0008|IMP-0007
SAT-CMD-004	Command Replay Detected	command	Anomaly	high	production	1	ASPIS Content	enabled	no	EX-0001.01|EX-0001
SAT-CMD-005	Command Flooding / High Command Rate	command	Anomaly	medium	production	1	ASPIS Content	enabled	no	EX-0013.01|EX-0013
SAT-CMD-006	Command Authentication Failure Spike	command	TTP	medium	production	1	ASPIS Content	enabled	no	IA-0008.01
SAT-CMD-007	Unencrypted Command on a Protected Link	command	TTP	medium	production	1	ASPIS Content	enabled	no	EX-0006|DE-0003.07
SAT-CMD-008	Rejected Command Counter Increasing	command	Anomaly	medium	production	1	ASPIS Content	enabled	no	EX-0013.02
SAT-CMD-009	First-Seen Command Type for Spacecraft	command	Anomaly	low	production	1	ASPIS Content	enabled	no	EX-0005.02
SAT-CMD-010	Privileged Command Without Prior Privilege Elevation	command	Correlation	high	production	1	ASPIS Content	enabled	no	IA-0007.02|PER-0005
SAT-FSW-001	Flight Software Application Stopped	fsw	TTP	high	production	1	ASPIS Content	enabled	no	DE-0001|IMP-0013
SAT-FSW-002	Flight Software Application Loaded or Reloaded	fsw	TTP	high	production	1	ASPIS Content	enabled	no	EX-0009.01|PER-0002.02
SAT-FSW-003	Onboard Log or Critical Data Store Cleared	fsw	TTP	high	production	1	ASPIS Content	enabled	no	DE-0010|IMP-0010
SAT-FSW-004	Event Reporting Disabled on Spacecraft	fsw	TTP	high	production	1	ASPIS Content	enabled	no	DE-0001|DE-0003.13
SAT-FSW-005	Onboard Memory Write / Poke	fsw	TTP	high	production	1	ASPIS Content	enabled	no	EX-0012.03|PER-0001
SAT-FSW-006	Onboard Integrity Check Failure (Checksum Miscompare)	fsw	Anomaly	high	production	1	ASPIS Content	enabled	no	PER-0001|EX-0009.01
SAT-FSW-007	Checksum / Integrity Monitoring Disabled	fsw	TTP	high	production	1	ASPIS Content	enabled	no	DE-0001|DE-0006
SAT-FSW-008	Unexpected Spacecraft Reset or Reboot	fsw	Anomaly	medium	production	1	ASPIS Content	enabled	no	IMP-0013|IMP-0008
SAT-FSW-009	Reboot Loop Detected	fsw	Anomaly	high	production	1	ASPIS Content	enabled	no	IMP-0013
SAT-FSW-010	Fault Management or Limit Checker Disabled	fsw	TTP	high	production	1	ASPIS Content	enabled	no	DE-0001
SAT-FSW-011	Onboard Table Load and Activation	fsw	TTP	medium	production	1	ASPIS Content	enabled	no	IMP-0009|EX-0012.04

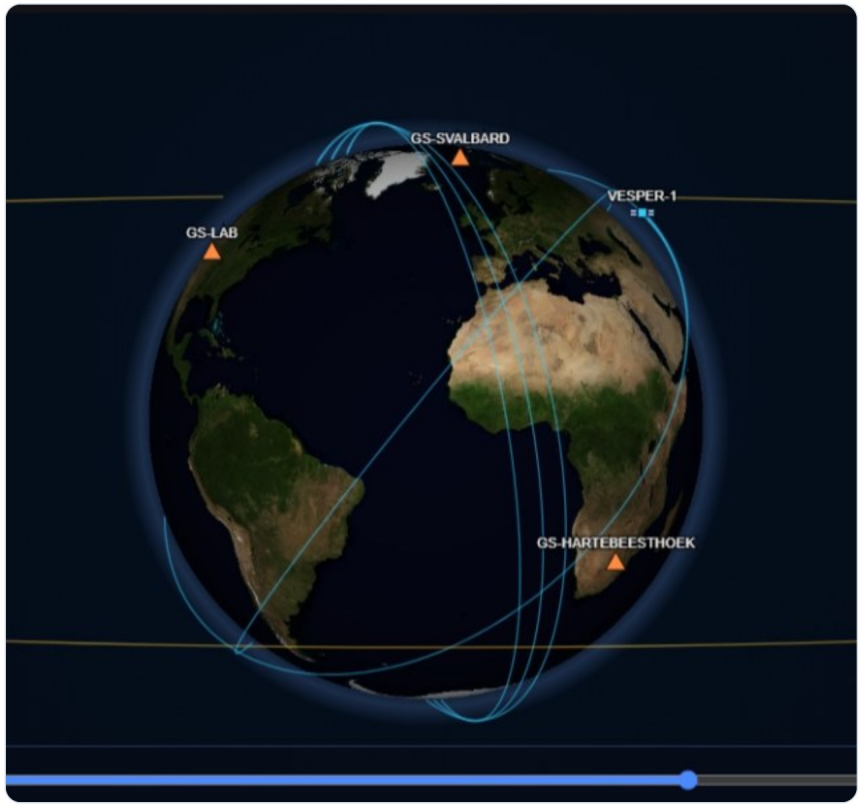
ON SCREEN

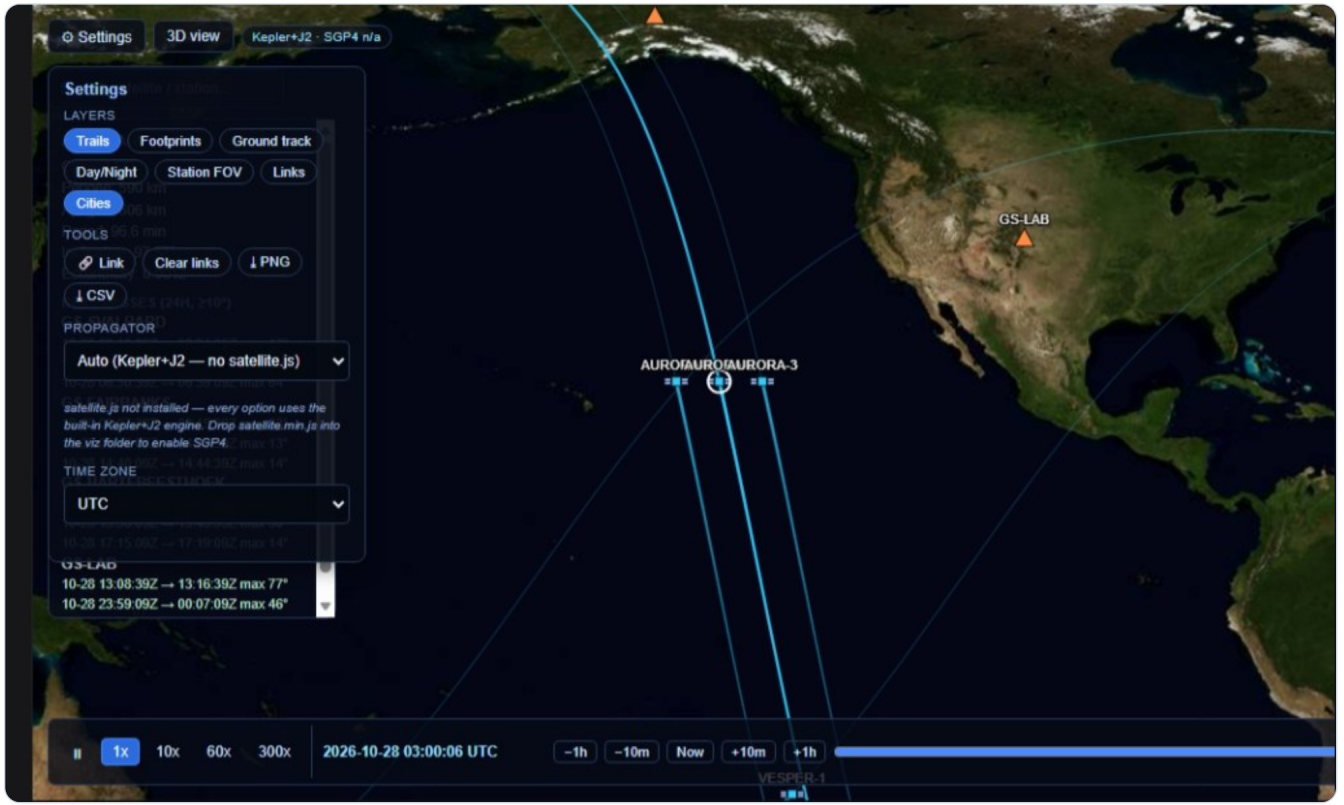
- Detections, enabled, disabled, experimental, overrides
- Catalog: domain, type, severity, status, version, state
- Enable or disable a detection with a reason

BEHIND IT

- Content-as-code: YAML compiled into saved searches
- Overrides stored in satsec_content_overrides
- Click a row to load it into the editor

Orbital View





ON SCREEN

- 3D globe or 2D map, switched with one click
- Click an object for orbit details and next passes
- Layers: trails, footprints, tracks, day/night, links
- Time bar: pause to 300×, jump ±1 h, scrub any time

BEHIND IT

- Orbits computed in the browser from TLEs (Kepler + J2)
- Contact links and threat arcs come from ASPIS data
- Fully offline: embedded imagery, air-gap ready

Content Versioning

Content inventory (versioned items)

detection_id	name	domain	type	severity	status	version	author
SAT-CMD-001	Spacecraft Command From Unauthorized Source	command	TTP	high	production	1	ASPIS Content
SAT-CMD-002	Spacecraft Command Outside Scheduled Contact Window	command	TTP	high	production	1	ASPIS Content
SAT-CMD-003	Hazardous Spacecraft Command Issued	command	TTP	high	production	1	ASPIS Content
SAT-CMD-004	Command Replay Detected	command	Anomaly	high	production	1	ASPIS Content
SAT-CMD-005	Command Flooding / High Command Rate	command	Anomaly	medium	production	1	ASPIS Content
SAT-CMD-006	Command Authentication Failure Spike	command	TTP	medium	production	1	ASPIS Content
SAT-CMD-007	Unencrypted Command on a Protected Link	command	TTP	medium	production	1	ASPIS Content
SAT-CMD-008	Rejected Command Counter Increasing	command	Anomaly	medium	production	1	ASPIS Content
SAT-CMD-009	First-Seen Command Type for Spacecraft	command	Anomaly	low	production	1	ASPIS Content
SAT-CMD-010	Privileged Command Without Prior Privilege Elevation	command	Correlation	high	production	1	ASPIS Content
SAT-FSW-001	Flight Software Application Stopped	fsw	TTP	high	production	1	ASPIS Content
SAT-FSW-002	Flight Software Application Loaded or Reloaded	fsw	TTP	high	production	1	ASPIS Content
SAT-FSW-003	Onboard Log or Critical Data Store Cleared	fsw	TTP	high	production	1	ASPIS Content
SAT-FSW-004	Event Reporting Disabled on Spacecraft	fsw	TTP	high	production	1	ASPIS Content
SAT-FSW-005	Onboard Memory Write / Poke	fsw	TTP	high	production	1	ASPIS Content
SAT-FSW-006	Onboard Integrity Check Failure (Checksum Miscompare)	fsw	Anomaly	high	production	1	ASPIS Content
SAT-FSW-007	Checksum / Integrity Monitoring Disabled	fsw	TTP	high	production	1	ASPIS Content
SAT-FSW-008	Unexpected Spacecraft Reset or Reboot	fsw	Anomaly	medium	production	1	ASPIS Content
SAT-FSW-009	Reboot Loop Detected	fsw	Anomaly	high	production	1	ASPIS Content
SAT-FSW-010	Fault Management or Limit Checker Disabled	fsw	TTP	high	production	1	ASPIS Content
SAT-FSW-011	Onboard Table Load and Activation	fsw	TTP	medium	production	1	ASPIS Content
SAT-FSW-012	Table Load Built for Different Spacecraft or Processor	fsw	Anomaly	high	production	1	ASPIS Content
SAT-FSW-013	Onboard System Time Changed	fsw	TTP	high	production	1	ASPIS Content
SAT-FSW-014	Onboard File Deletion	fsw	TTP	medium	production	1	ASPIS Content
SAT-FSW-015	Invalid Command Code / Length Error Burst	fsw	Anomaly	medium	production	1	ASPIS Content

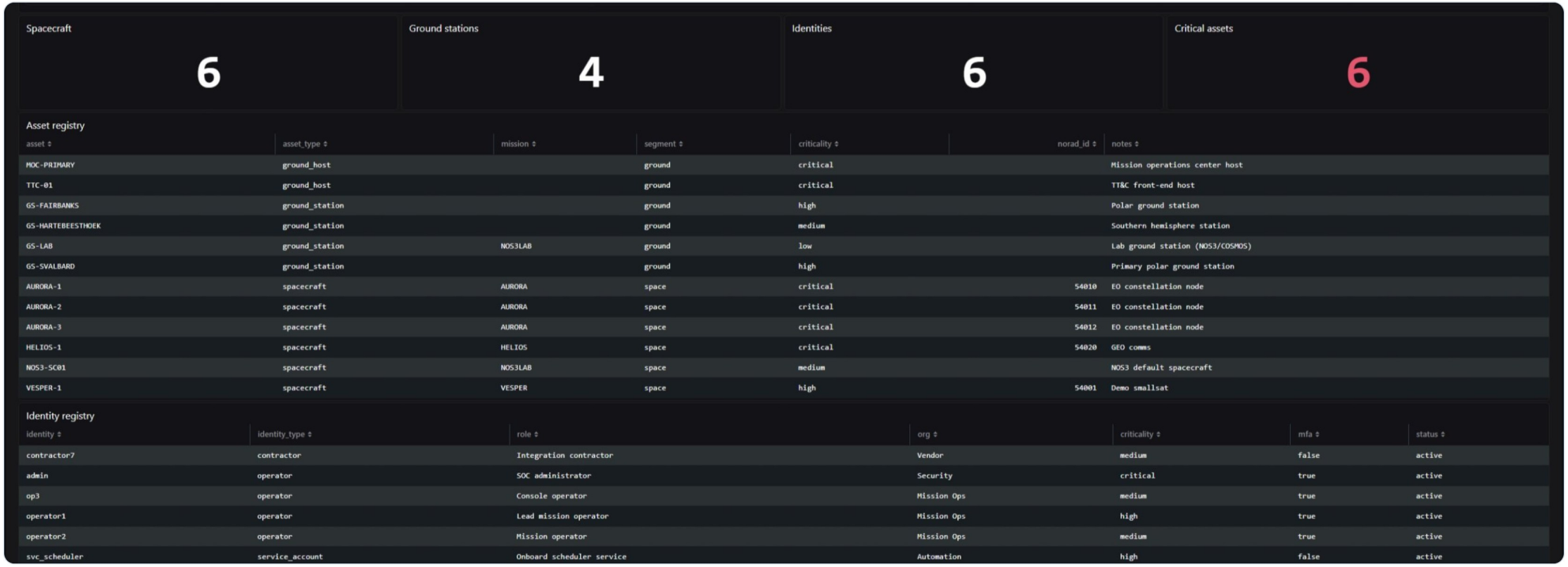
ON SCREEN

- Every detection with its domain, type and severity
- Status, deployed revision and author per detection
- The page also charts revisions and keeps a change log

BEHIND IT

- Each detection carries its revision and author in YAML
- The inventory is built from the content files
- The ES Content Versioning equivalent

Asset & Identity



ON SCREEN

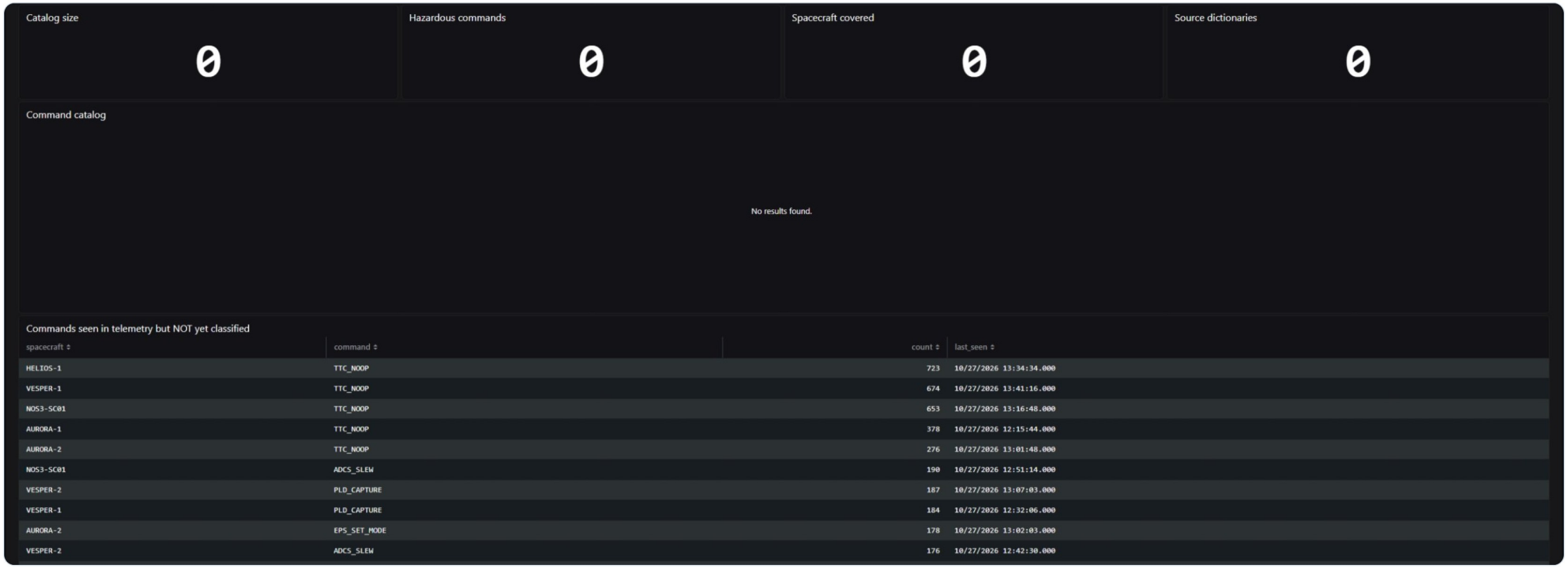
- Spacecraft, ground stations, identities, critical assets
- Asset registry: mission, segment, criticality, NORAD ID
- Identity registry and an add or update form

BEHIND IT

- satsec_assets and satsec_identities KV collections
- Criticality feeds urgency and every risk score
- Demo assets and identities load from seed CSVs

Command Catalog

Empty by design until your flight dictionary is imported



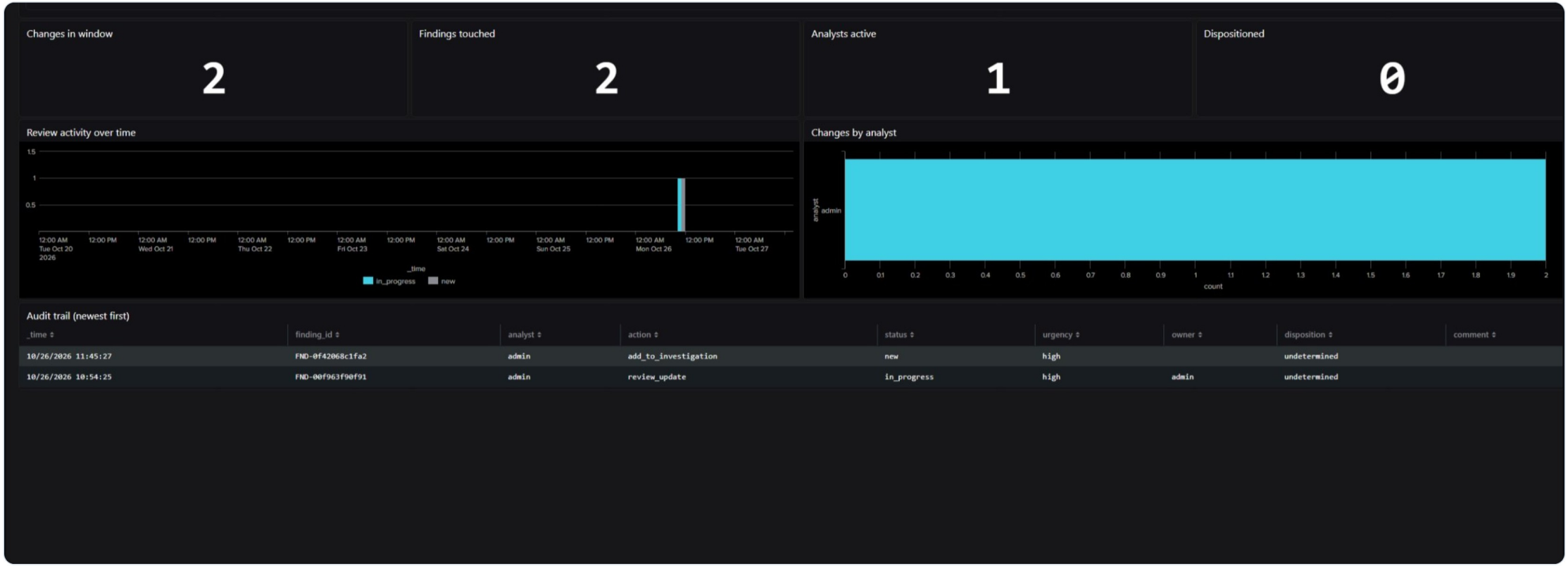
ON SCREEN

- Catalog size, hazardous commands, spacecraft, dictionaries
- Catalog and commands seen but not yet classified
- Single-command editor and import instructions

BEHIND IT

- satsec_command_catalog KV classifies each mnemonic
- Hazard levels come from your XTCE or COSMOS dictionary
- Drives SAT-CMD-003 (hazardous) and -010 (privileged)

Incident Review Audit



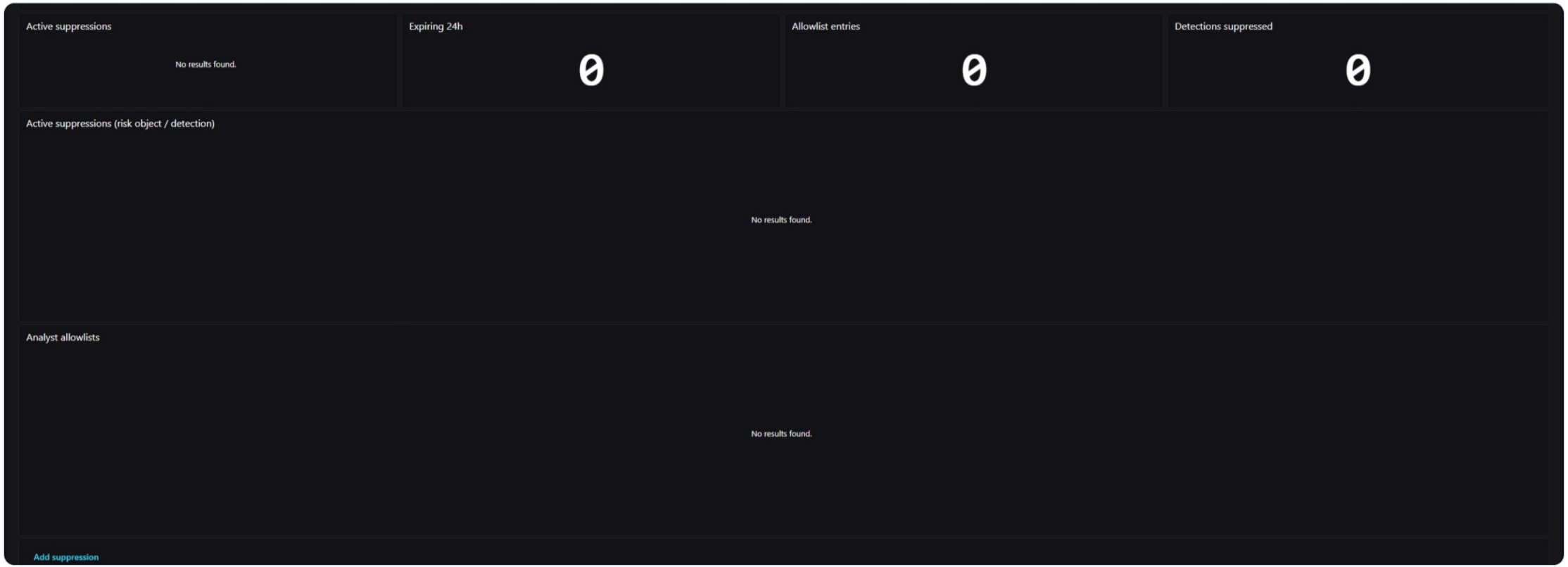
ON SCREEN

- Changes, findings touched, analysts, dispositioned
- Review activity over time and changes by analyst
- Audit trail: who, when, action, new state

BEHIND IT

- Append-only satsec_finding_review_audit KV collection
- Written by Incident Review on every change
- The ES incident review audit equivalent

Suppression Audit



ON SCREEN

- Active, expiring in 24 h, allowlist, detections suppressed
- Suppressions by risk object and detection; allowlists
- Add a suppression for a number of hours

BEHIND IT

- satsec_suppressions and satsec_allowlist KV collections
- Every finding rule applies the suppression macro first
- The ES suppression audit equivalent

OPERATING ASPIS

Running it day to day

Install and upgrade

Manage Apps › Install app from file. An upgrade installs over the existing app, and browsers load the new code.

Demo mode

17 searches named “ASPIS Demo -” keep the lab populated. Disable them all before real telemetry arrives.

First run

Run “ASPIS Setup - Load Demo Missions” and “Load Demo Assets” once; coverage fills within the hour.

Connect telemetry

Adapters post to the “aspis” HEC token (off by default). Point the satsec_idx macros at existing indexes.

Permissions

Analysts need read on the app’s indexes and write on its KV collections; writes use their own session.

Quality gates

44 automated tests and a package validator run on every build, before anything ships.

COMPARED WITH ENTERPRISE SECURITY

Where ASPIS matches ES, and where it does not

ES capability	ASPIS	Status
Correlation searches	127 detections compiled from YAML content	Equivalent
Risk-based alerting	Risk index, 6 risk rules, mission and phase weights	Equivalent
Incident Review	ES layout: queues, timeline, bulk edit, audit	Equivalent
Investigations	List and workbench: timeline, artifacts, history	Equivalent
Content and versions	Content Management and Versioning consoles	Equivalent
Asset and identity	Registries plus two entity investigators	Partial
Threat intelligence	Indicator KV, match detection, risk factor, import	Lighter
Adaptive response	Three actions: outbox, add intel, webhook	Partial
Mission Control extras	AI disposition, SOAR playbooks, co-editing	Not built

KNOWN LIMITS

What to know before production

Finding volume	The risk-threshold rule re-raises a busy object every hour. Tune the thresholds and throttle window per site.
Status permissions	ES limits status transitions per role; in ASPIS anyone with KV write access can move a finding.
Legacy forms	Five registry forms still write with outputlookup; a scoped setting keeps the risky-command prompt off.
Lab data only	Every number in this deck is synthetic demo data on a lab server, not a real mission.

One place to see, score and work every threat to the mission

01

Install ASPIS and load the asset and identity registries

02

Disable demo jobs; connect real telemetry

03

Import the flight dictionary into the catalog

04

Tune risk thresholds and urgency for the fleet